

IT 320 Course Project
Semester-2, 1447H



Software Product Release

NAR'AK

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Table of Content

CHAPTER 1: INTRODUCTION	4
1.1 THE PROBLEM	5
1.2 THE SOLUTION	6
1.3 THE PRODUCT	7
1.3.1 Product Vision	7
1.3.2 Product Roadmap	8
1.3.3 Objectives.....	8
1.3.4 Scope.....	10
1.4 THE SCRUM TEAM.....	11
2 CHAPTER 2: DOMAIN ANALYSIS	11
2.1 LITERATURE REVIEW	11
3 CHAPTER 3: REQUIREMENTS ENGINEERING	13
3.1 REQUIREMENTS ELICITATION AND ANALYSIS	13
3.1.1 Interviews.....	14
3.1.2 Questionnaire.....	17
3.2 SYSTEM USERS.....	20
3.3 USE CASE DIAGRAM.....	22
3.4 PRODUCT BACKLOG	23
4 CHAPTER 4: SYSTEM DESIGN AND DEVELOPMENT	34
4.1 SYSTEM ARCHITECTURE.....	34
4.2 CLASS DIAGRAM	36
4.3 DATA DESIGN.....	37
4.4 COMPONENT DESIGN	44
4.5 INTERFACE DESIGN	48
5 CHAPTER 5: TESTING.....	49
5.1 USER STORY ACCEPTANCE TESTING	49
5.2 INTEGRATION TESTING.....	79
5.3 USER ACCEPTANCE TESTING.....	85
5.4 DISCUSSION	89
6 CHAPTER 6: CONCLUSION AND FUTURE WORK.....	91
6.1 GLOBAL AND LOCAL IMPACT.....	91
6.2 PROBLEMS AND CHALLENGES ENCOUNTERED DURING THE SOFTWARE DEVELOPMENT	92
6.3 LIMITATIONS OF THE SYSTEM.	93
6.4 THE MAIN CONTRIBUTION OF THE PROJECT	93
6.5 FUTURE WORK.	94
7 REFERENCES.....	94
8 APPENDIX.....	95
8.1 APPENDIX A – INTERVIEW	95
8.1.1 interview questions	95
8.1.2 interview transcripts.....	96
8.2 APPENDIX B – QUESTIONNAIRES	111
8.2.1 Questionnaires questions	111

8.2.2 Complete user responses	112
8.3 APPENDIX C – UAT QUESTIONNAIRES	113
8.3.1 Questionnaires Graphs results	113
8.3.2 Questionnaires Questions.....	116

Table of Tables

Table 1- Scrum Team.....	11
Table 2- Competing Websites and their Features	13
Table 3- Interviewee information	14
Table 4-Product Backlog	32
Table 5- Data Dictionary of Entities	39
Table 6- Data Dictionary of relationships.....	39
Table 7- Attributes Description	43
Table 8 - User Story Acceptance Testing	79
Table 9 - Integration Scenario 1.....	83
Table 10 - Integration Scenario 2.....	85
Table 11- Interview 1 Transcript	96
Table 12- Interview 2 Transcript	100
Table 13- Interview 3 Transcript	104
Table 14- Interview 4 Transcript	108

Table of Figures

Figure 1-Roadmap	8
Figure 2- Questionnaire Q1 result.....	18
Figure 3- Questionnaire Q2 result.....	18
Figure 4- Questionnaire Q3 result.....	18
Figure 5- Questionnaire Q4 result.....	19
Figure 6- Questionnaire Q5 result.....	19
Figure 7- Use Case Diagram.....	22
Figure 8-Jira ScreenShot1	33
Figure 9-Jira ScreenShot2.....	33
Figure 10- Nar'ak System Architecture	35
Figure 11- Nar'ak Class Diagram.....	36

Figure 12- Nar'ak ER Diagram	37
Figure 13- Booking Flowchart.....	45
Figure 14- Upload Results Flowchart	46
Figure 15- Block Flowchart	47
Figure 16- Nar'ak Sitemap	48
Figure 17 - Admin Integration hierarchy	80
Figure 18 - Lab Integration hierarchy	80
Figure 19 - Customer Integration hierarchy	81
Figure 20 - Gender	86
Figure 21 - Age Group	87
Figure 22 - Occupation	87
Figure 23 - Technical Experience	88
Figure 24	113
Figure 25	113
Figure 26	113
Figure 27	114
Figure 28	114
Figure 29	114
Figure 30	115
Figure 31	115
Figure 32	115
Figure 33	116

Chapter 1: Introduction

Health and wellbeing services are increasingly connected to technology, as many people now depend on digital platforms to access essential services. One important area within this field is laboratory testing, which plays a key role in monitoring health and supporting health awareness. With the growth of digital health services, users expect more convenient and organized ways to request services and access information.

However, accessing laboratory services and understanding test results is not always easy. Users may face difficulties when requesting laboratory tests, accessing their results online, or understanding medical reports due to complex medical terminology. These challenges can reduce health awareness and limit individuals' ability to follow up on their health information effectively.

Using technology in health-related services is especially important at both the global and local levels. Globally, digital health platforms have become essential in improving access to healthcare services, increasing health awareness, and supporting more efficient use of healthcare resources. At the local level, in Riyadh, such technologies can improve access to laboratory services, support health awareness, and help reduce pressure on hospitals and clinics. This project is limited to Riyadh and does not aim to operate on a national or global scale, allowing the system to focus on local laboratories, local users, and realistic healthcare practices within the city.

This document presents the Nar'ak project and provides an overview of the context and motivation behind the system. It outlines the main problem addressed by the project, explains the overall direction and scope, and introduces the key elements that guide its development. These include the product vision, roadmap, project objectives, defined scope, and the roles of the team members involved in building the platform.

1.1 The Problem

Laboratory testing is an essential part of healthcare, as it supports health monitoring and early detection of potential health issues. However, in Riyadh, accessing laboratory services is not always easy for all individuals. Many users face challenges related to physical access, time constraints, and the effort required to complete laboratory testing procedures, especially when services are fragmented across different providers.

A clear real-world example of this problem occurs when a user needs to perform Vitamin D and HbA1c tests. The user may contact several laboratories separately to compare availability and services, then arrange a home visit through phone calls or messaging applications such as WhatsApp. After the tests are completed, results are often sent as PDF files or text messages that are difficult to interpret.

When the user repeats the same tests months later, there is no simple way to compare the new results with previous ones, as the reports are stored across different laboratories and in different formats.

This project focuses on the challenges related to accessing laboratory services and managing laboratory test information within Riyadh. These challenges include heavy traffic congestion, long travel times to hospitals and laboratories, and delays in receiving test results due to the high volume of tests processed daily in major healthcare facilities. Such conditions make follow-up difficult for users and reduce the overall efficiency of accessing laboratory services. By narrowing the focus to these issues, the project addresses a specific and well-defined problem within the local healthcare context of Riyadh.

1.2 The Solution

This project proposes a web-based health and wellbeing platform that acts as an intermediary between individuals and certified laboratories in Riyadh. The platform aims to simplify access to laboratory services by providing a centralized system through which customers can request laboratory tests, including home sample collection, without the need to communicate separately with multiple providers.

Through the platform, laboratories are able to manage test requests and upload laboratory reports directly to customers' accounts. Test results are presented in a clear and organized format, allowing customers to easily access their reports from different laboratories in one place. In addition, the platform supports tracking and comparing previous laboratory results over time, helping customers monitor changes in their health data.

The solution contributes on improving accessibility, organization, and understanding of laboratory test information. It provides educational explanations and general health guidance related to test results.

1.3 The Product

1.3.1 Product Vision

For individuals in Riyadh who are unable to visit laboratories in person due to personal circumstances, health conditions, mobility limitations, or time constraints.

Who faces difficulties in arranging laboratory tests, accessing their results, and managing reports from multiple laboratories.

The Nar'ak platform *is a* web-based health and wellbeing platform,

That enables customers to request laboratory tests with home sample collection, compare services from multiple certified laboratories, and access and organize their test results in one place for easier follow-up

Unlike individual laboratory platforms such as Sulaiman al Habib Medical Group [\[1\]](#) and Al Borg Medical Laboratories[\[2\]](#), which operate as standalone service providers,

Our product acts as a centralized intermediary that allows customers to compare and choose between multiple laboratories through a single platform, while supporting health awareness without providing medical diagnosis or treatment.

1.3.2 Product Roadmap

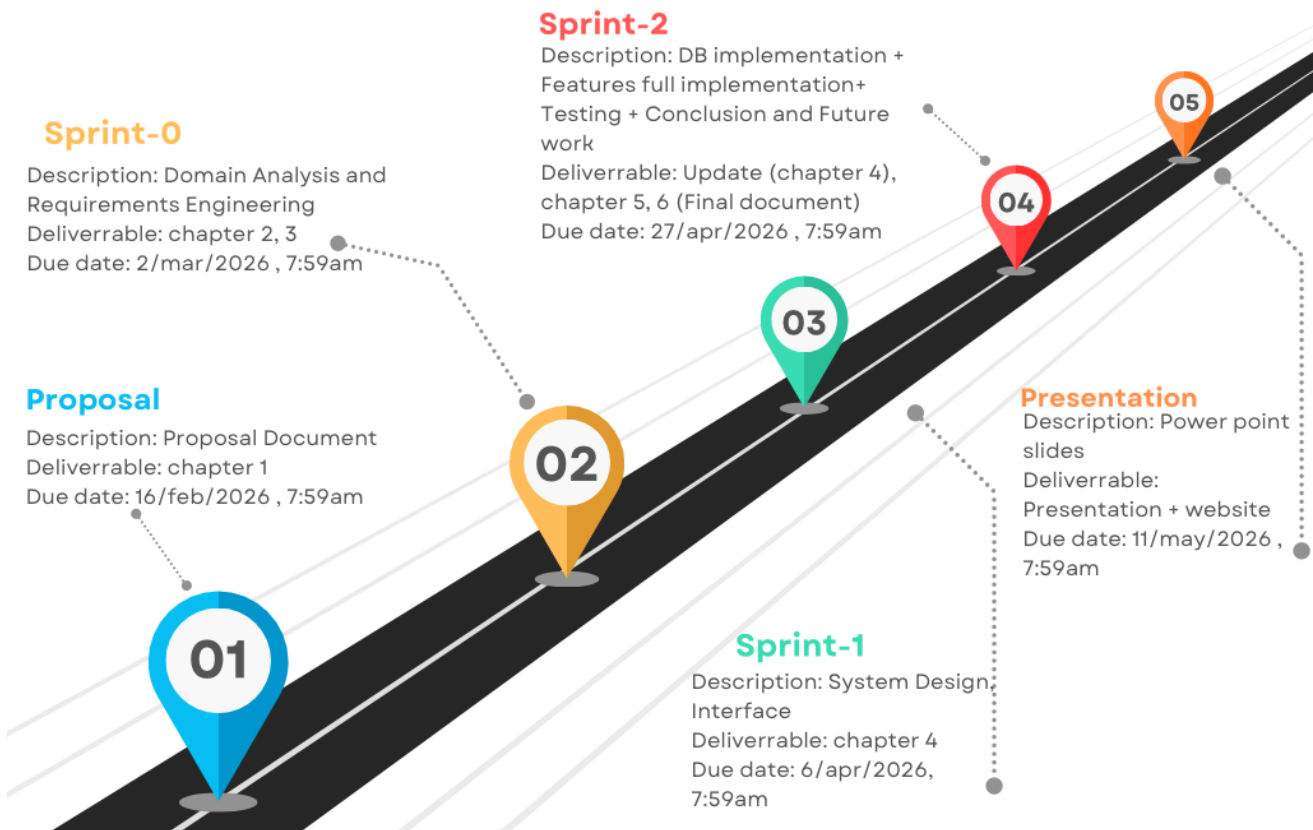


Figure 1-Roadmap

1.3.3 Objectives

1. Product

These objectives outline the main ways in which the product provides value to customers, administrators, and laboratories through its core functionalities, which include the following:

Customer objectives

- The customer can register securely to create a personal account in the system.
- The customer can log in securely to access their account.
- The customer can log out securely to end the session safely.
- The customer can browse available laboratories and their available test types.
- The customer can add the desired laboratory, test type, and available appointment time.

- The customer can edit their appointments.
- The customer can delete their appointments.
- The customer can view the scheduled appointment details.
- The customer can search laboratory tests.
- The customer can track the status of submitted requests.
- The customer can access laboratory reports through a single account.
- The customer can access and view laboratory test results in a clear and user-friendly format that is easy to understand.
- The customer can view compared current and previous laboratory results over time.

Admin Objectives (Platform Management)

- The admin can log in securely to access their account.
- The admin can log out securely to end the session safely.
- The admin can manage customer and laboratory accounts.
- The admin can monitor laboratory test appointments submitted by customers and all there data.
- The admin can ensure smooth coordination and workflow between customers and laboratories.

Laboratory Objectives (System Role)

- The laboratory can log in securely to access their account.
- The laboratory can log out securely to end the session safely.
- The laboratory can add the available slots to the system monthly
- The laboratory can receive test appointments selected by customers.
- The laboratory can upload laboratory reports within the system.
- The laboratory can update the progress of the appointment requests.

2. Project

To successfully complete the Nar'ak project, the team will:

- Study customer needs in Riyadh and identify challenges in accessing laboratory services.
- Collect and analyze data related to laboratories and system requirements.

- Design the system architecture, database, and user interfaces.
- Develop a functional web-based platform that connects customers with certified laboratories.
- Implement features for test requests and home sample collection.
- Create a system that allows laboratories to upload reports directly to customers' accounts.
- Conduct User Acceptance Testing (UAT) to evaluate usability.
- Improve the system based on user feedback before final delivery.

3. Learning

Through this project, the team will gain new learning experiences including:

- Practicing working within a scrum team and task coordination using agile-based project workflows.
- Improving skills in collaborative development and version control within a group setting by using GitHub tool, Jira.
- Understanding how real-world systems are planned, structured, and developed.
- Gaining basic awareness of laboratory blood tests, such as common test names and result formats, to better align the system with user needs.
- Learning new programming languages like PHP.
- Learning how to integrate and work with database in our website to manage information securely and efficiently.

1.3.4 Scope

The scope of the Nar'ak project defines the high-level boundaries of the system at this stage of development. The platform is designed to operate only within the city of Riyadh, and all supported laboratories are limited to laboratories located in Riyadh.

The system targets customers of all age groups; however, only customers aged 18 years and above are allowed to request laboratory tests for themselves. Parents or legal guardians are permitted to request laboratory tests on behalf of their children.

The platform supports Arabic language only. Nar'ak is implemented as a web-based application and is accessible only through laptop or desktop using Google Chrome [3] web browser. Mobile phones and tablets are out of scope. The system is optimized for screens with a minimum resolution of 1440px.

The scope of the project is limited to interactions between customers and laboratories only. The system does not support doctors as users, does not integrate with insurance providers or governmental health systems, and does not provide medical diagnosis, treatment, or clinical decision-making. Laboratory test data is presented for viewing and follow-up purposes only.

1.4 The Scrum Team

Scrum Team	
Product Owner:	Noora Hussain Alsaiani
Developers:	Noora Hussain Alsaiani Yara Abdullah Zakzouk Norah Nasser Al Hussain Farah Ali Alhamed
Scrum Master (SM):	
Stakeholders:	

Table 1- Scrum Team

2 Chapter 2: Domain Analysis

2.1 Literature Review

Before developing Nar'ak, a market analysis was conducted to identify existing platforms that provide laboratory-related services in Riyadh. The purpose of this review is to analyze competitors, understand their strengths and limitations, and identify opportunities for improvement.

Competitor #1: Dr. Sulaiman Al Habib Medical Group[1]

Dr. Sulaiman Al Habib Medical Group provides both a website and a mobile application that allow users to book medical appointments and access various healthcare services, including laboratory testing.

Strength:

- 1) Provides access to laboratory test results through a secure user account.
- 2) Offers online appointment booking for laboratory and other medical services.
- 3) Integrates additional healthcare services such as radiology and specialized clinics within the same platform.

Limitation:

- 1) Integrates additional healthcare services such as radiology and specialized clinics within the same platform.
- 2) The search functionality is limited and may not effectively filter or locate specific laboratory tests.

Competitor #2: Al Borg Diagnostics[\[2\]](#)

Al Borg Diagnostics provides both a website and a mobile application that allow users to book laboratory appointments and access certain healthcare services, including home consultations.

Strength:

- 1) Sends SMS notifications to inform users when laboratory test results are ready.
- 2) Offers teleconsultation services, allowing users to communicate with healthcare professionals remotely.
- 3) Provides online booking options for laboratory services.

Limitation:

- 1) Does not allow users to compare laboratory test results across different laboratories.
- 2) Laboratory results are typically provided as downloadable PDF files, which may limit interactive tracking or visualization of data over time.

Feature	NAR'AK	Dr Suliman Al Habib	Al Borg Diagnostics
Booking home tests online	✓	✓	✓
Centralized report storage	✓	✗	✗
Application system	✗	✓	✓
Result comparison	✓	✗	✗

Table 2- Competing Websites and their Features

Nar'ak differs from competing platforms by focusing specifically on centralized laboratory services within one unified system. Unlike Dr. Sulaiman Al Habib and Al Borg Diagnostics, which operate as standalone healthcare providers, Nar'ak acts as an intermediary that connects users to multiple laboratories in one place. It uniquely offers centralized report storage and result comparison over time, allowing users to track their laboratory results across different providers, a feature not supported by the competitors.

3 Chapter 3: Requirements Engineering

3.1 Requirements Elicitation and Analysis

To gather the requirements for the NAR'AK system, we used multiple requirements elicitation techniques to better understand user needs and expectations. First, we conducted user interviews with individuals who have experience booking laboratory tests, which helped us identify common problems such as difficulty finding available appointments, challenges in understanding medical reports, and issues accessing previous results. We also distributed questionnaires to collect feedback from a larger group of users to validate the findings from the interviews. In addition, we observed how users currently book laboratory tests, whether through phone calls, hospital apps, or in-person visits, which allowed us to understand the existing workflow and its inefficiencies. We identified key stakeholders, including patients, laboratories, and system administrators, and considered their different needs when defining the requirements. Furthermore, we analyzed similar healthcare systems such as Dr. Sulaiman Al Habib and Al Borg Diagnostics laboratories to understand common features like appointment booking, report access, and result display, and to identify areas for improvement such as automatic result comparison and centralized report storage. By combining these methods, we were able to define both functional and non-functional requirements that accurately reflect the needs of the NAR'AK users.

3.1.1 Interviews

- Interviewee information

This section we interviewed 4 participants representing different age groups, occupations, and technical backgrounds in Riyadh.

Interviewee #	Age	Gender	Nationality	Location	Occupation	Tech level	Main challenge
1	28	Male	Saudi Arabia	Riyadh	Employee	High	Busy work schedule and difficulty finding suitable appointment times.
2	21	Female	Saudi Arabia	Riyadh	University student	High	University schedule conflicts and rush-hour traffic.
3	45	Female	Saudi Arabia	Riyadh	Housewife	Moderate	Heavy traffic and long distance.
4	60	Female	Saudi Arabia	Riyadh	Housewife	Moderate	Confusion when reviewing results.

Table 3- Interviewee information

- Interviews results and findings

This section presents an analysis of the interview responses collected from the participants. The full interview questions and transcripts are provided in the [Appendices](#). The purpose of this analysis is to identify common challenges, behavioral patterns, and user needs related to booking laboratory tests, accessing laboratory reports, and visiting laboratories in Riyadh. The findings from these interviews are used to derive system requirements and define key functional priorities for the Nar'ak platform.

Question 1:

Finding:

Across the four responses, booking-related challenges were strongly associated with time constraints, limited appointment availability, and fragmented communication with laboratories.

- **Interviewee 1** employee emphasized scheduling conflicts due to work responsibilities and limited appointment windows.
- **Interviewee 2** highlighted conflicts between morning classes and laboratory availability, which forces her to attend during rush hours.
- **Interviewee 3** described the booking process as exhausting due to the need to check multiple laboratories and wait for responses.
- **Interviewee 4** participant mentioned that early appointments require rearranging her daily responsibilities.

Pattern Analysis:

The responses show a consistent issue related to scheduling conflicts and limited appointment flexibility. Participants from different backgrounds struggle to align laboratory appointments with work, university, or family responsibilities. The main challenge appears to be time coordination.

Question 2:

Findings:

All participants reported difficulty understanding laboratory reports, though the degree of confusion varied.

- Younger participants (**Interviewee 1 and Interviewee 2**) stated they understand basic numbers but struggle with medical abbreviations and abnormal ranges.
- The older participants (**Interviewee 3 and Interviewee 4**) expressed emotional discomfort and uncertainty when reviewing reports.
- Several participants mentioned that slight deviations from normal values cause confusion.
- The formatting differences between laboratories add to the difficulty of interpretation.

Pattern Analysis:

All participants reported difficulty understanding laboratory reports. The problem is not accessing the reports, but interpreting medical terminology, abbreviations, and reference ranges. This indicates a gap between data availability and user comprehension.

Question 3:

Findings:

All participants stated that they keep their previous test results. However, none reported having them stored in an organized or easily comparable format.

- Results are stored as PDFs, emails, printed copies, or WhatsApp files.
- Different laboratories use different formats and layouts.
- Manual comparison is required to identify trends over time.
- Some participants avoid reviewing old results due to inconvenience.

Pattern Analysis:

Participants keep their previous test results, but they are stored in scattered formats. Comparing results over time requires manual effort, making long-term tracking inconvenient and inconsistent.

Question 4:

Findings:

Traffic congestion in Riyadh emerged as one of the most consistently mentioned problems across all demographics.

- The 28-year-old employee (**Interviewee 1**) struggles with leaving work and dealing with peak traffic.
- The 21-year-old student (**Interviewee 2**) must attend during rush hours due to morning classes.
- The 45-year-old housewife (**Interviewee 3**) described the experience as stressful and exhausting.
- Another 60-year-old participant (**Interviewee 4**) reported feeling more tired from sitting in the car than from the test itself.

Pattern Analysis:

Traffic congestion in Riyadh affects all participants regardless of age or occupation. Travel time, peak hours, and parking difficulties increase stress and negatively impact the overall testing experience.

Question 5:

Findings:

All participants expressed that home sample collection is important, but for different reasons.

- **Interviewee 1** values time efficiency.
- **Interviewee 2** values flexibility around university commitments.
- **Interviewee 3** values reduced stress and easier management of family duties.
- **Interviewee 4** associates home collection with emotional comfort.

Pattern Analysis:

Convenience is a shared priority among all participants. Home sample collection is valued because it reduces travel, saves time, and minimizes stress across different lifestyles.

3.1.2 Questionnaire

To validate the interview findings and gather broader insights, a questionnaire was conducted to collect quantitative data from a larger group of users. The survey focused on challenges related to booking laboratory tests, understanding reports, and accessing laboratory services in Riyadh.

The questionnaire was created using Google Forms and distributed through WhatsApp and social media. Only participants aged 18 and above were included. The survey contained mainly closed-ended questions, and the results were analyzed and presented in graphical form. The full questionnaire and detailed responses are provided in the [Appendix](#).

Questionnaire graphs result:

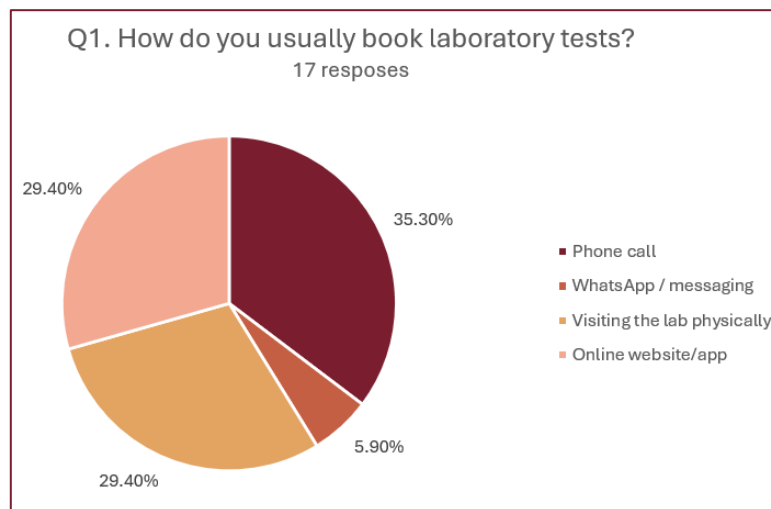


Figure 2- Questionnaire Q1 result

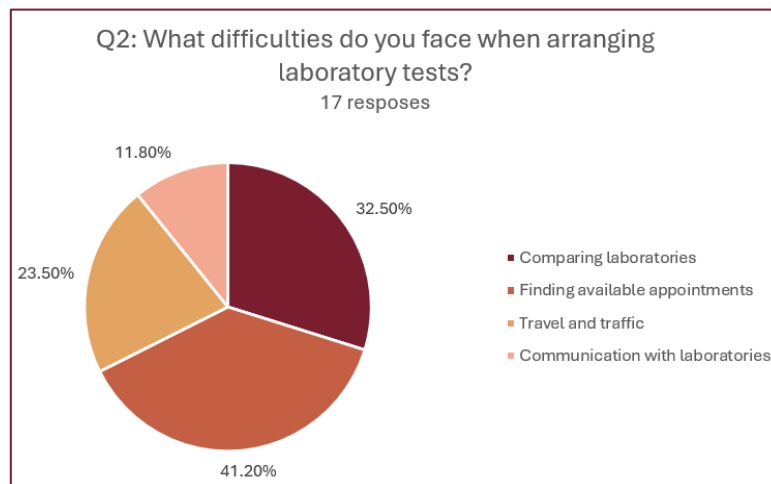


Figure 3- Questionnaire Q2 result

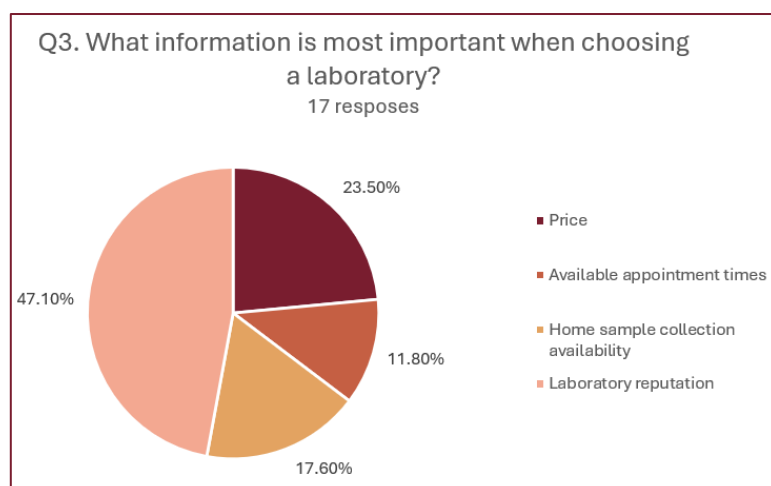


Figure 4- Questionnaire Q3 result

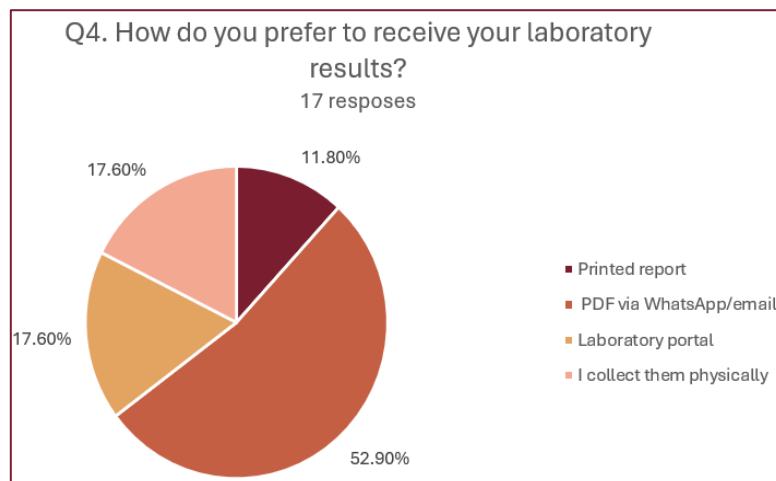


Figure 5- Questionnaire Q4 result

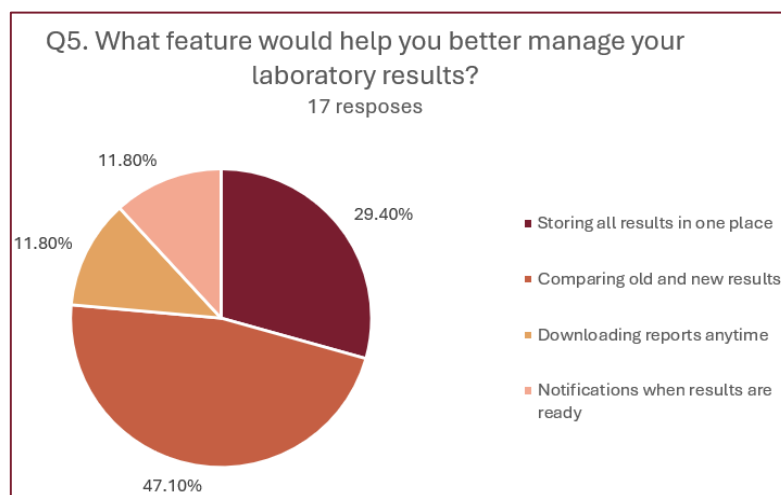


Figure 6- Questionnaire Q5 result

Summary of findings:

The analysis of both the interviews and the questionnaire responses revealed consistent and complementary insights regarding users' experiences with laboratory services in Riyadh.

From the interviews, participants emphasized scheduling conflicts and limited flexibility when booking appointments. Working individuals struggled to align appointments with work hours, students were forced to book during rush hours due to morning classes, and older participants described the booking process as exhausting. These qualitative findings are supported by the questionnaire results, where 41.2% of respondents identified difficulty finding available appointment times as their main challenge, and 32.5% reported difficulty comparing laboratories.

Regarding laboratory reports, interview participants expressed uncertainty and lack of confidence when interpreting medical terminology and numerical values. Several described feeling anxious when reviewing results without explanation. This concern is reinforced by the questionnaire findings, where 47.1% of respondents selected “comparing old and new results” as the most helpful feature, and 29.4% emphasized the importance of storing all results in one place. This demonstrates a clear need for improved result management and tracking.

Traffic and travel difficulties were strongly highlighted during interviews, where participants described traffic congestion as stressful and time-consuming. Similarly, the questionnaire showed that 23.5% of respondents experience travel and traffic as a key difficulty when arranging tests.

Finally, decision-making factors were identified through the questionnaire, with 47.1% prioritizing laboratory reputation, while price (23.5%) and home sample collection availability (17.6%) were also considered important. Interview responses further showed that convenience and reduced stress significantly influence user preferences.

Overall, combining qualitative interview insights with quantitative questionnaire data provides a comprehensive understanding of user needs. The findings indicate the necessity for improved appointment visibility, centralized report storage, easier comparison of results over time, and reduced reliance on physical visits. These conclusions directly guided the functional requirements and feature design of the Nar’ak platform.

3.2 System Users

The Nar’ak platform targets individuals residing in Riyadh who require laboratory testing services and seek a more convenient and organized way to access laboratory tests and results. The system serves three primary user groups: Customers, Laboratories, and the System Administrator.

For customers:

- Customers should be residents of Riyadh, including Saudis and non-Saudis living in the city.
- Customers must be at least 18 years old to request laboratory tests for themselves. Parents or legal guardians may request tests on behalf of their children.

- There are no specific educational requirements; customers may have diverse educational backgrounds. However, the system must present information in a clear and simple format to ensure understanding.
- Customers are not required to have medical knowledge, as most of them are not healthcare professionals and may have limited understanding of laboratory terminology.
- Customers should have basic technical skills to navigate the website, create an account, log in, and manage appointments using the Google Chrome browser [\[3\]](#).
- Customers are expected to have general familiarity with online websites and digital services.
- Some customers may face mobility limitations, health conditions, or time constraints due to traffic congestion in Riyadh, which increases the importance of home sample collection services.

For laboratories:

- Laboratories must be certified and located within Riyadh.
- Laboratory staff are expected to have professional knowledge in laboratory testing and report management.
- Laboratory staff should have high technical skills to manage appointments, upload reports, and update request statuses through the system.

For the system administrator:

- The administrator should have high technical knowledge to manage customer and laboratory accounts.
- The administrator is responsible for monitoring system operations and ensuring smooth coordination between customers and laboratories.

3.3 Use Case Diagram

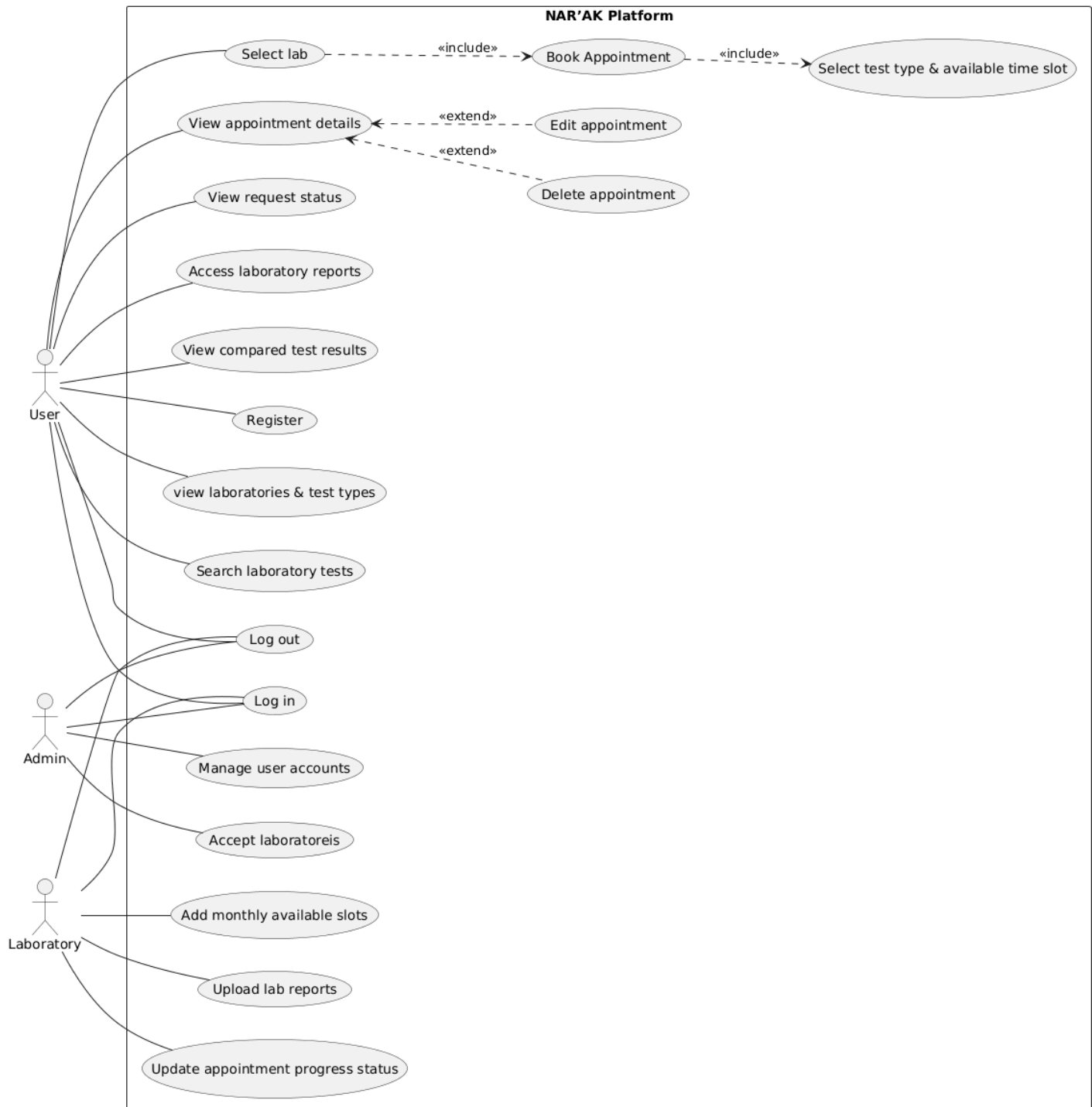


Figure 7- Use Case Diagram

3.4 Product Backlog

PBI User Story	Size Story Points	Type	Acceptance Criteria
As a customer, I want to register in the system so that I can create my personal account.	3	Functional Feature	- As a customer, if any required field is empty, I should see an error message and the account should not be created. -As a customer, if I enter an invalid email format, the system should show an error message. -As a customer, if the email or phone number already exists, the system should show a message that the account already exists. -As a customer, when I type my password, the password characters should be masked (hidden). -As a customer, after successful registration, I should be redirected to my dashboard.
As a NAR'AK user, I want to log in securely so that I can access my dashboard and services.	2	Functional Feature	-As a user, if I enter correct email and password and click "Login", I should access my dashboard successfully. -As a user, if I enter incorrect email or password, I should see an error message. -As a user, if I leave email or password empty, I should see an error message.

			-As a user, when I type my password, the characters should be masked (hidden).
As a NAR'AK user, I want to log out so that my account remains secure after using the system.	1	Functional Feature	-As a user, if I click the "Logout" button, I should be logged out successfully. -As a user, after logging out, I should be redirected to the login page.
As a customer, I want to browse available laboratories so that I can see which labs are registered in the platform.	3	Functional Feature	-As a customer, when I view my dashboard, I should see a list of available laboratories. -As a customer, each laboratory should display its basic information.
As a customer, I want to view available test types for each laboratory so that I can choose the correct medical test.	3	Functional Feature	-As a customer, when I open a laboratory page, I should see the available test types for that lab.
As a customer, I want to search for a test type so that I can find what I need quickly.	3	Functional Feature	-As a customer, if I type a test name in the search bar, I should see matching results. -As a customer, if no matching test is found, I should see a "No results found" message.
As a customer, I want to select a laboratory so that I can book an appointment.	3	Functional Feature	-As a customer, I should be able to select one laboratory.

			-As a customer, I should not continue booking without selecting a laboratory. -As a customer, after selecting a laboratory, I should see the available test types and time slots.
As a customer, I want to select the type of test so that I can request the correct medical examination.	2	Functional Feature	-As a customer, after selecting a laboratory, I should see the available test types. -As a customer, I should be able to select multiple test types. -As a customer, I should not continue booking without selecting a test type.
As a customer, I want to select an available time slot so that I can schedule my appointment at a convenient time.	3	Functional Feature	-As a customer, after selecting a lab, I should see the available time slots. -As a customer, I should be able to select one available time slot. -As a customer, I should not continue booking without selecting a time slot.
As a customer, I want to view my appointment details so that I can confirm the booking information.	3	Functional Feature	-As a customer, after selecting the lab, test type, and time slot, I should see an appointment summary page. -As a customer, I should be able to confirm the booking only after viewing the summary. -As a customer, if I click “Confirm”, the appointment should appear in my appointments list successfully.

			-As a customer, after successful booking, I should see a confirmation message. -As a customer, if I click “Cancel” on the summary, the booking should not be created.
As a customer, I want to edit my appointment so that I can change the time or test if needed.	5	Functional Feature	-As a customer, if I click “Edit”, I should be able to change the test type or time slot. -As a customer, if I select a new available slot and save, the appointment should update successfully. -As a customer, after editing, I should see a confirmation message
As a customer, I want to cancel/delete my appointment so that I can remove it if I no longer need it.	3	Functional Feature	-As a customer, if I click “Cancel”, I should be asked to confirm the cancellation. -As a customer, if I confirm, the appointment status should change to “Cancelled”. -As a customer, after cancellation, I should see a confirmation message.
As a customer, I want to track the status of my test request so that I know the progress of my appointment.	5	Functional Feature	-As a customer, when I open my appointment details, I should see the current status of the appointment. -As a customer, the status should clearly indicate whether it is Pending,

			Confirmed, In Progress, Completed, or Cancelled. -As a customer, if the laboratory updates the status, I should see the updated status when I refresh or reopen the page.
As a customer, I want to access all my laboratory reports in one account so that I can manage my medical history.	5	Functional Feature	-As a customer, when I open the Reports section, I should see a list of all my previous reports. -As a customer, if I click on a report, it should open successfully.
As a customer, I want to view my test results in a clear and understandable format so that I can easily read them	3	Functional Feature	-As a customer, when I open a report, the test name and date should be clearly displayed. -As a customer, the result values should be clearly labeled. -As a customer, the results should not appear in a confusing or unorganized format.
As a customer, I want to view a comparison of my current and previous test results so that I can monitor changes over time.	5	Functional Feature	-As a customer, when a new test result is available, the system should automatically compare it with my previous result of the same test type. -As a customer, I should see both the latest and previous result displayed clearly.

			-As a customer, the difference between the results should be visible.
As a laboratory representative, I want to add available appointment slots monthly so that customers can book home sample collection at convenient times.	3	Functional Feature	-As a laboratory representative, if I open the availability section, I should be able to select available dates using a calendar interface instead of entering the date manually. -As a laboratory representative, if I attempt to select a past date from the calendar, I should be prevented from doing so. -As a laboratory representative, if I successfully add a date and time slot, it should appear immediately in the system and be visible to users for booking.
As a laboratory representative, I want to receive test appointments selected by customers so that I can prepare for sample collection and manage my schedule efficiently.	5	Functional Feature	-As a laboratory representative, if a customer books an appointment using one of my available slots, I should see the appointment appear in my laboratory appointment list. -As a laboratory representative, if an appointment is booked by a customer, the selected slot should automatically become unavailable and should not be displayed as available to other customers. -As a laboratory representative, if I open an appointment details page, I

			should see the booking information including customer details, test type, appointment date, and appointment time. -As a laboratory representative, if there are no booked appointments, I should see a clear message indicating that there are currently no appointments.
As a laboratory representative, I want to upload laboratory reports within the system so that customers can access their test results from their personal accounts.	3	Functional Feature	-As a laboratory representative, if I select a specific booked appointment and open the results form, I should be able to enter the test results for that appointment. -As a laboratory representative, if I submit the results form with all required fields filled correctly, the results should be saved successfully and linked to the selected appointment. -As a laboratory representative, if I leave any required result field empty and submit, I should see an error message asking me to fill in all required fields. -As a laboratory representative, if I enter a numeric test result in a text-only format when the test requires a number, I should see an error message indicating that the result must be numeric.

			-As a laboratory representative, if I save the results successfully, the appointment status should automatically update to “Completed”.
As a laboratory representative, I want to update the progress status of appointment requests so that customers can track the current stage of their laboratory test.	3	Functional Feature	-As a laboratory representative, if I open an appointment request, I should be able to update its status to one of the allowed progress stages (e.g., Pending, Confirmed, In Progress, Completed, or Cancelled.). -As a laboratory representative, if I update the appointment status and save it, the new status should be stored successfully and reflected immediately in the laboratory dashboard. -As a laboratory representative, if I update the appointment status, the customer should be able to view the updated status in their account when tracking the request.
As an admin, I want to view and manage customer and laboratory accounts so that I can maintain platform integrity and ensure only verified accounts operate on the system.	5	Functional Feature	-As an admin, when I view my dashboard, I should see a list of latest appointments, reported customers and laboratories. -As an admin, I can delete/add a laboratory account.

			-As an admin, I should be able to block reported customer account. -As an admin, if I block a customer account, the customer should not be able to log in.
As an admin, I want to monitor all laboratory test requests and view their detailed information so that I can ensure smooth workflow and resolve issues when needed.	8	Functional Feature	-As an admin, when I view the request section in my dashboard, I should see a list of all laboratory test requests. -As an admin, if I click on a specific request, I should be able to view its detailed information. -As an admin, if no requests exist, I should see a clear message indicating that there are no requests available. -As an admin, if a laboratory updates the request status, I should see the updated status immediately when refreshing or reopening the page. -As an admin, I should be able to identify delayed or pending requests based on their status and appointment date.
As NAR'AK users, I want the interface to be simple and easy to understand so that I can navigate the system without confusion.	3	Nonfunctional feature	- As a Nar'ak user, if I access my dashboard, I should clearly understand the main purpose of the platform within a few seconds. - As a Nar'ak user, if I navigate between pages, I should find consistent

			layout, colors, and button styles throughout the system. - As a Nar'ak user, if I click on any button or link, I should immediately understand its function based on its label. - As a Nar'ak user, if I make an error while filling a form, I should see a clear and understandable error message.
As a Nar'ak user, I want the system to protect all personal and laboratory data so that my medical information remains secure and confidential.	5	Nonfunctional feature	- As a Nar'ak user, I want my password to be stored securely so that it is not visible or stored in plain text within the system. - As a Nar'ak user, I want access to laboratory results and personal information to require authentication so that unauthorized individuals cannot view sensitive data. - As a Nar'ak user, I want the system to enforce proper access control so that only authorized roles can access specific information.

Table 4-Product Backlog

Jira screenshot:

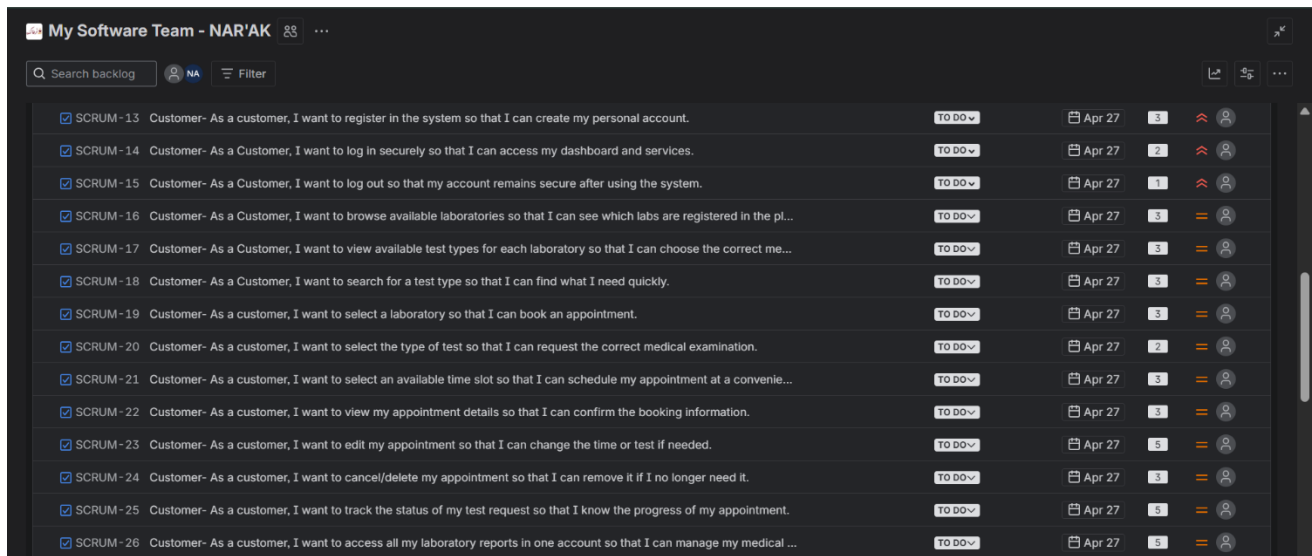


Figure 8-Jira ScreenShot1

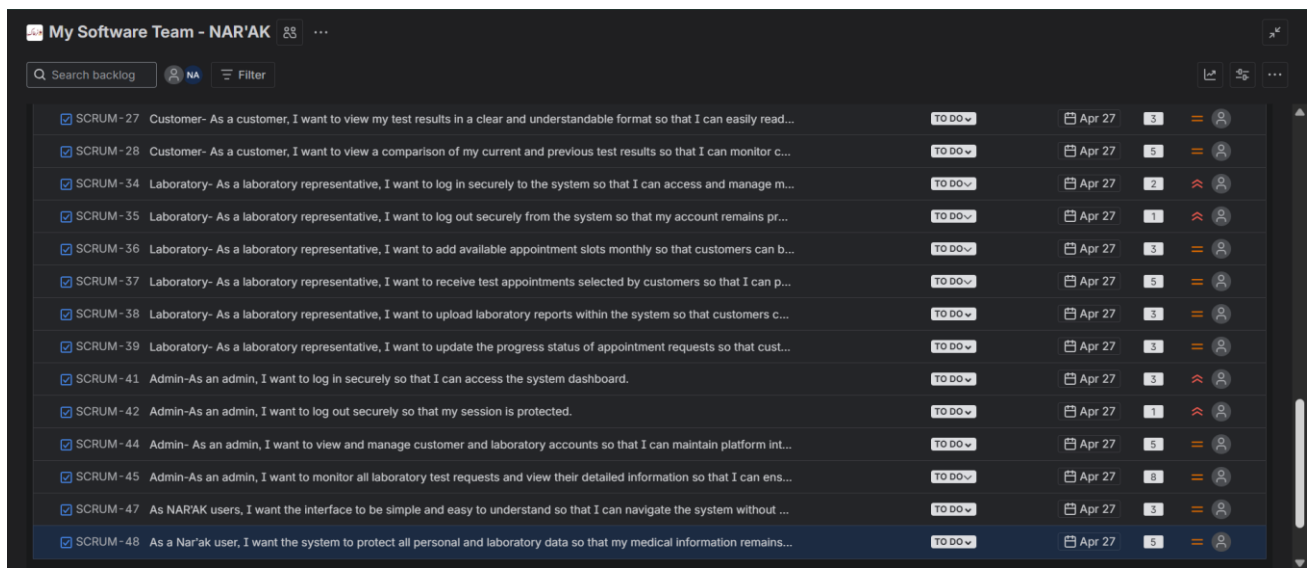


Figure 9-Jira ScreenShot2

[Jira link](#)

4 Chapter 4: System Design and Development

4.1 System Architecture

This section presents the system architecture selected for the Nar'ak platform. The system architecture defines how different components interact to deliver the required functionality. It directly affects the system's performance, scalability, maintainability, and overall organization.

After analyzing different architectural approaches, the Client-Server Architecture was selected for the Nar'ak system. This architecture divides the system into two main components: the client-side and the server-side. This separation allows efficient communication between users and the system while ensuring proper handling of data and processing also improving system organization and allowing easier maintenance and updates. Additionally, this architecture supports scalability by allowing multiple clients to interact with the server simultaneously. It also enhances security by centralizing data and system logic on the server, ensuring controlled access to sensitive information.

The Client-Server Architecture is well-suited for the Nar'ak system as it ensures efficient communication between components, supports system scalability, and provides a secure and organized structure for handling user requests and data management.

The client-side represents the user interface through which users interact with the system. In Nar'ak, the system supports three main types of users: customers, laboratories, and administrators.

This component includes all web pages such as customer dashboard (registration, login, browsing laboratories, selecting test types, booking appointments, viewing reports, and comparing test results), laboratories dashboard (receive appointments, upload test results, add available time slots and report customers) and admin dashboard (add new laboratory and block reported customers). It provides a user-friendly experience by displaying information clearly, capturing user input, and allowing smooth navigation through the system.

The client-side is implemented using web technologies such as HTML, CSS, and JS, and is accessed through a web browser.

The server-side is responsible for handling all system processing and data management. It includes both the business logic and the database.

This component performs all core system operations such as:

- Authentication and authorization of users
- Managing bookings, appointments, and available time slots
- Handling laboratory reports and test results
- Managing user accounts and system administration
- Searching for laboratory tests and tracking request status

The server processes user requests, performs validation, executes system logic, and communicates with the database to store and retrieve data. The system uses a MySQL database to store all relevant data, including but not limited to customers, laboratories, test types, appointments, time slots, and test results.

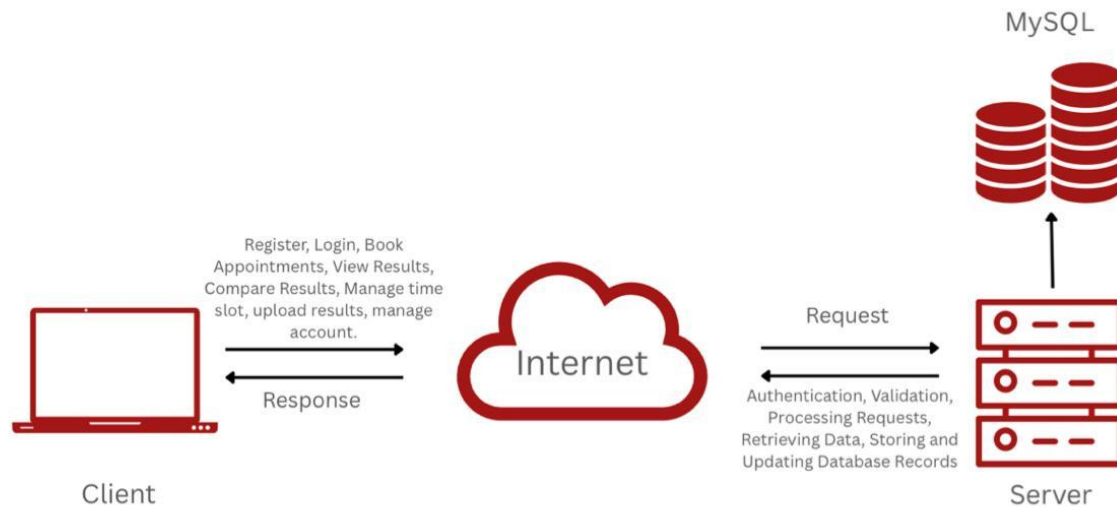


Figure 10- Nar'ak System Architecture

Figure 10 illustrates the Client-Server Architecture of the Nar'ak platform. Users interact with the system through the client-side interface, which sends requests through the Internet to the server-side.

The server processes requests, performs validation and system operations, and communicates with the MySQL database to store and retrieve data.

4.2 Class Diagram

This section presents the class diagram of the Nar'ak system. The class diagram provides a clear visualization of the system structure by illustrating the main classes, along with their attributes, methods, and relationships. It helps in understanding how the system is organized and how its components interact with each other. The diagram also provides a structured overview of the system design before implementation.

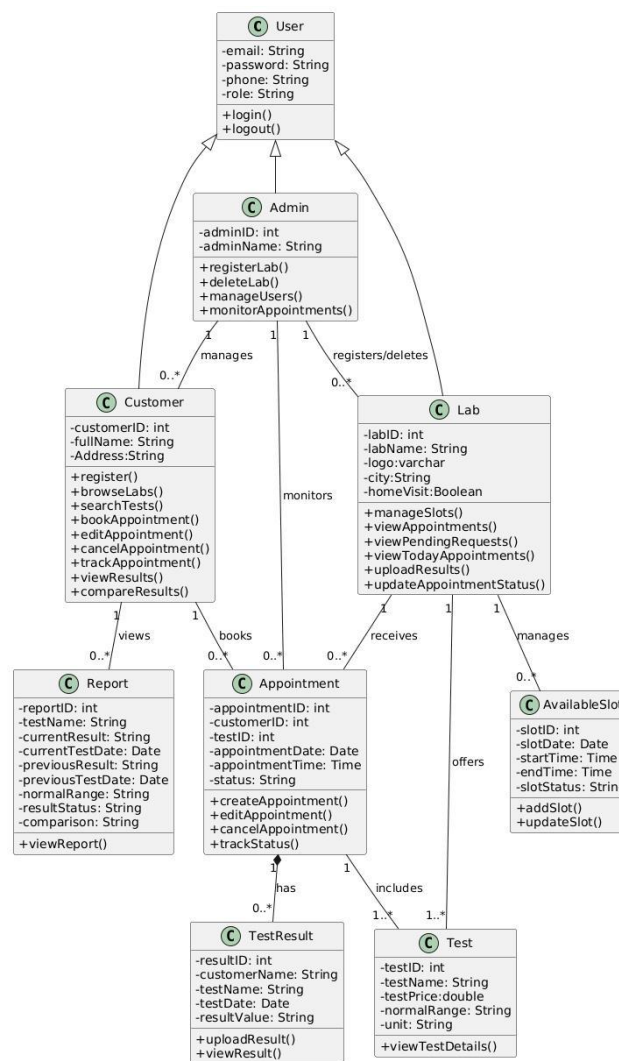


Figure 11- Nar'ak Class Diagram

4.3 Data Design

This section presents the data design of the Nar'ak System, which is built upon a relational SQL database. The database is structured to efficiently manage the core entities of the system. The design follows standard relational modeling principles, ensuring data integrity through the use of primary keys, foreign keys, and normalized table structures that eliminate redundancy and maintain consistency across the system.

The data design is presented in three parts:

First, the Entity Relationship (ER) diagram provides a visual representation of the entities and the relationships between them, illustrating how the different components of the system interact.

Second, the relational schema formally defines each table, its attributes, and the constraints applied, including primary and foreign key references.

Third, the data dictionary offers a comprehensive description of every entity, relationship, and attribute in the database, serving as a complete reference for understanding the structure and content of the database.

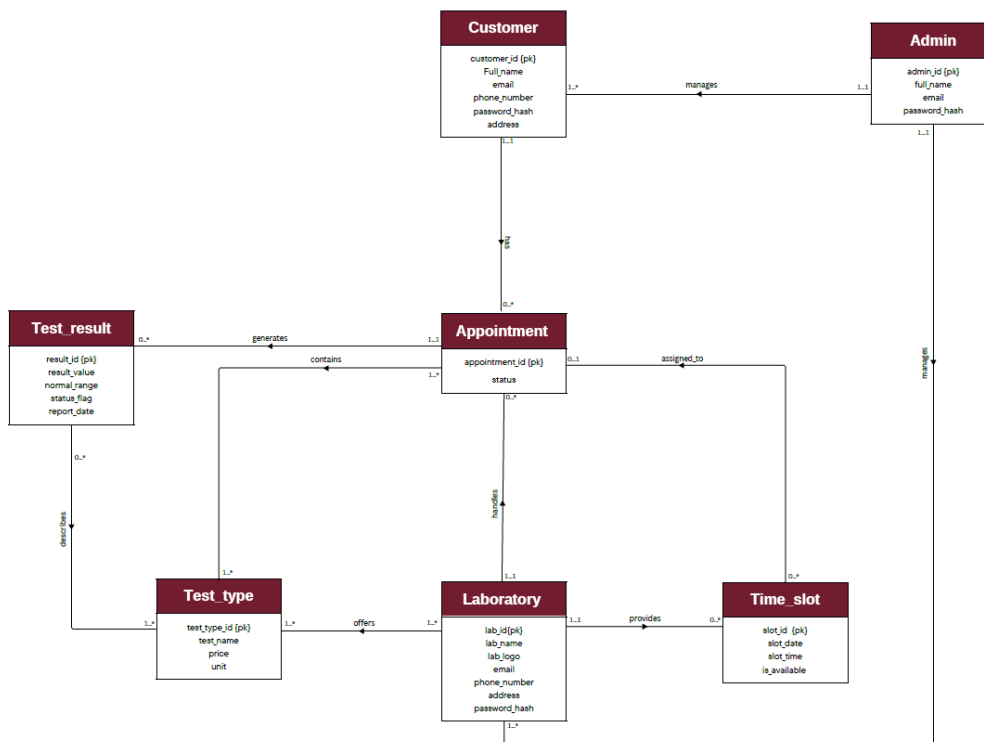


Figure 12- Nar'ak ER Diagram

Relational Schema:

Customer (customer_id, Full_name, email, phone_number, password_hash, address)

Primary Key: customer_id

Laboratory (lab_id, lab_name, lab_logo, email, phone_number, address, password_hash)

Primary Key: lab_id

Admin (admin_id, full_name, email, password_hash)

Primary Key: admin_id

Time_slot (slot_id, slot_date, slot_time, is_available, lab_id)

Primary Key: slot_id

Foreign Key: lab_id References Laboratory (lab_id)

Test_type (test_type_id, test_name, price, unit, lab_id)

Primary Key: test_type_id

Foreign Key: lab_id References Laboratory (lab_id)

Appointment (appointment_id, customer_id, lab_id, slot_id, status)

Primary Key: appointment_id

Foreign Key: customer_id References Customer (customer_id)

Foreign Key: lab_id References Laboratory (lab_id)

Foreign Key: slot_id References Time_slot (slot_id)

Test_result (result_id, result_value, normal_range, status_flag, report_date, appointment_id, test_type_id)

Primary Key: ResultID

Foreign Key: appointment_id References Appointment (appointment_id)

Foreign Key: test_type_id References Test_type (test_type_id)

Data Dictionary:

Data Dictionary showing description of all entities.

Entity Name	Description	Occurrence
Customer	A Customer refers to any person who makes use of our lab appointment system, which are the individuals who are interested in booking a lab test from a registered laboratory.	A customer can have none or many appointments.
Laboratory	A Laboratory, within the context of our application, is a facility that offers medical tests that is available for appointments.	A laboratory handles none or many appointments. A laboratory provides none or many time slots. A laboratory offers one or many test types.
Appointment	An appointment in our application represents that one of the time slots of a laboratory has been booked by a customer.	An appointment generates none or many test results. An appointment contains one to many test types.
Admin	An Admin is a privileged user who manages customers and laboratories on the platform.	An admin manages one or more customers. An admin manages one or more laboratories.
Time_slot	A Time_slot represents an available date and time window offered by a laboratory during which a customer may book an appointment.	A time slot is assigned to none or one appointment.
Test_type	A Test_type is a catalogue entry for a specific medical test offered by a laboratory, including its price and unit of measurement.	A test type is offered by one or many laboratories.
Test_result	A Test_result stores the outcome of a specific test performed during an appointment, including the measured value and its interpretation.	A test result describes one to many test types.

Table 5- Data Dictionary of Entities

Data Dictionary showing description of all relationships:

Entity Name	Multiplicity	Relationship	Entity Name	Multiplicity
Customer	1..1	Has	Appointment	0..*
Laboratory	1..1	Handles	Appointment	0..*
Laboratory	1..1	Provides	Time_slot	0..*
Laboratory	1..*	Offers	Test_type	1..*
Time_slot	0..*	Assigned_to	Appointment	0..1
Admin	1..1	Manages	Customer	1..*
Admin	1..1	Manages	Laboratory	1..*
Appointment	1..1	Generates	Test_result	0..*
Appointment	1..*	Contains	Test_type	1..*
Test_result	0..*	Describes	Test_type	1..*

Table 6- Data Dictionary of relationships

Data Dictionary showing description of all attributes:

Entity Name	Attribute	Description	Data Type	Length	Nulls	Multi-Valued	Default Value	Range	PK
Customer	customer_id	Unique id to identify the customer	VARCHAR	5	N	N	N	00001-99999	Y
	Full_name	Customer's full name	String	30	N	N	N	N	N
	email	Customer's email address	VARCHAR	45	N	N	N	N	N
	phone_number	Customer's phone number	VARCHAR	10	N	N	N	N	N
	password_hash	Customer's password to sign in the system	VARCHAR	-	N	N	N	N	N
	address	Customer's address	VARCHAR	-	N	N	N	N	N
Laboratory	lab_id	Unique id to identify the laboratory	VARCHAR	3	N	N	N	001-999	Y
	lab_name	Lab's name	String	25	N	N	N	N	N
	lab_logo	The file path or URL pointing to the laboratory's official logo image,	VARCHAR	-	N	N	N	N	N

		used for display purposes across the platform							
	email	Lab's email address	VARC HAR	45	N	N	N	N	N
	phone_number	Lab's phone number	VARC HAR	10	N	N	N	N	N
	address	Lab's address	VARC HAR	-	N	N	N	N	N
	password_hash	Laboratory's password to sign in the system	VARC HAR	-	N	N	N	N	N
Appointment	appointment_id	Unique id to identify the appointment	VARC HAR	-	N	N	N	N	Y
	status	The current lifecycle state of the appointment booking, updated as the appointment progresses through the system	String	15	N	N	pending	pending/confirmed/in process/completed/canceled	N
Time_slot	slot_id	Unique id to identify each slot	VARC HAR	-	N	N	N	N	Y

	slot_date	Date of slot	Date	-	N	N	N	N	
	slot_time	Time of slot	Time	-	N	N	N	N	N
	is_available	A flag indicating whether this time slot is still open for booking	Boolean	-	N	N	TRUE	TRUE/FALSE	N
Test_type	test_type_id	Unique id to identify each test type	VARCHAR	-	N	N	N	N	Y
	test_name	Name for each test type	String	-	N	N	N	N	N
	price	Price for each test type	Decimal	-	N	N	N	>0.0	N
	unit	Measurement unit	String	10	N	N	N	N	N
Test_result	result_id	Unique id to identify each test result	VARCHAR	10	N	N	N	0000000001-9999999999	Y
	result_value	The actual measured outcome of the test performed	VARCHAR	100	N	N	N	N	N
	normal_range	used to interpret whether the result is	String	100	N	N	N	Changes based on test type	N

		within healthy bounds							
	status_flag	A classification label indicating how the result compares to the normal range	String	15	N	N	N	Normal/High/Low	N
	report_date	The date when the test result was finalized and made available to the customer	Date	-	N	N	N	N	N
Admin	admin_id	Unique id to identify the admin	VARCHAR	3	N	N	N	001-999	Y
	full_name	Admin's full name	String	100	N	N	N	N	N
	email	Admin's email address	String	150	N	N	N	N	N
	password_hash	Admin's password to enter the system	VARCHAR	-	N	N	N	N	N

Table 7- Attributes Description

4.4 Component Design

In this section, we present the design of three key components in the NARAK system. These components focus on the main system functionality rather than basic functions such as login, registration, or logout. For each component, we provide its definition, pseudocode, and activity flowchart logic.

- Booking an appointment from available slots (Customer Dashboard)

The customer can book an appointment by choosing a laboratory from the dashboard, then selecting one to three tests and one available date and time from the booking form. The form displays the selected laboratory information and only the available slots provided by that laboratory.

Pseudocode:

BEGIN

 DISPLAY "Dashboard Page"

 IF user clicks on "Book" button THEN

 NAVIGATE to "Booking Form"

 ENDIF

 DO

 INPUT selected tests, date, time

 IF selected tests IS EMPTY OR date IS EMPTY OR time IS EMPTY THEN

 DISPLAY "Error: All fields are required! Please fill in the missing fields."

 ENDIF

 WHILE selected tests IS EMPTY OR date IS EMPTY OR time IS EMPTY

 IF user clicks on "Back" button THEN

 NAVIGATE to "Dashboard Page"

 ENDIF

 IF user clicks on "Confirm Booking" button THEN

 Add appointment information (selected tests, date, time) to database

 DISPLAY message to confirm the booking.

 NAVIGATE to "Dashboard Page"

 ENDIF

END

Flowchart:

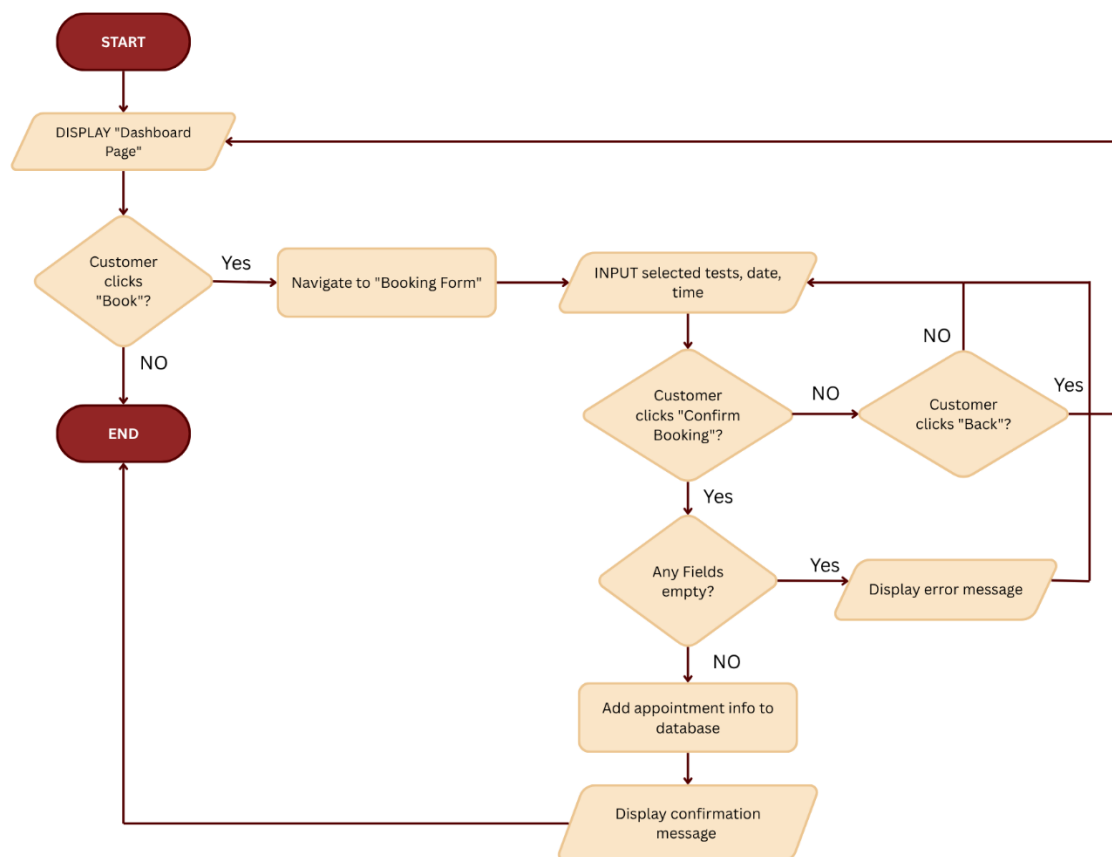


Figure 13- Booking Flowchart

- Upload test results (Laboratory Dashboard)

This function allows laboratory staff to upload test results for appointments that are in progress. The system displays a list of appointments waiting for results entry. When a lab selects an appointment, a form is shown with the booked tests, and the lab enters the corresponding result values. After submission, the system saves the results in the database and updates the appointment status to complete.

Pseudocode:

BEGIN

DISPLAY "Upload Results Page"

RETRIEVE appointments with status "In Progress" from database

DISPLAY list of "In Progress" appointments

IF lab selects an appointment from the list THEN

RETRIEVE booked tests for the selected appointment from database

DISPLAY result-entry form with input fields for the retrieved tests

ENDIF

DO

INPUT result values for each displayed test

IF any result value IS EMPTY THEN

DISPLAY "Error: All result fields are required! Please fill in the missing fields."

ENDIF

WHILE any result value IS EMPTY

IF user clicks on "Upload Results" button THEN

Add entered result values to database

Update appointment status to "Complete" in database

DISPLAY message to confirm successful upload.

ENDIF

END

Flowchart:

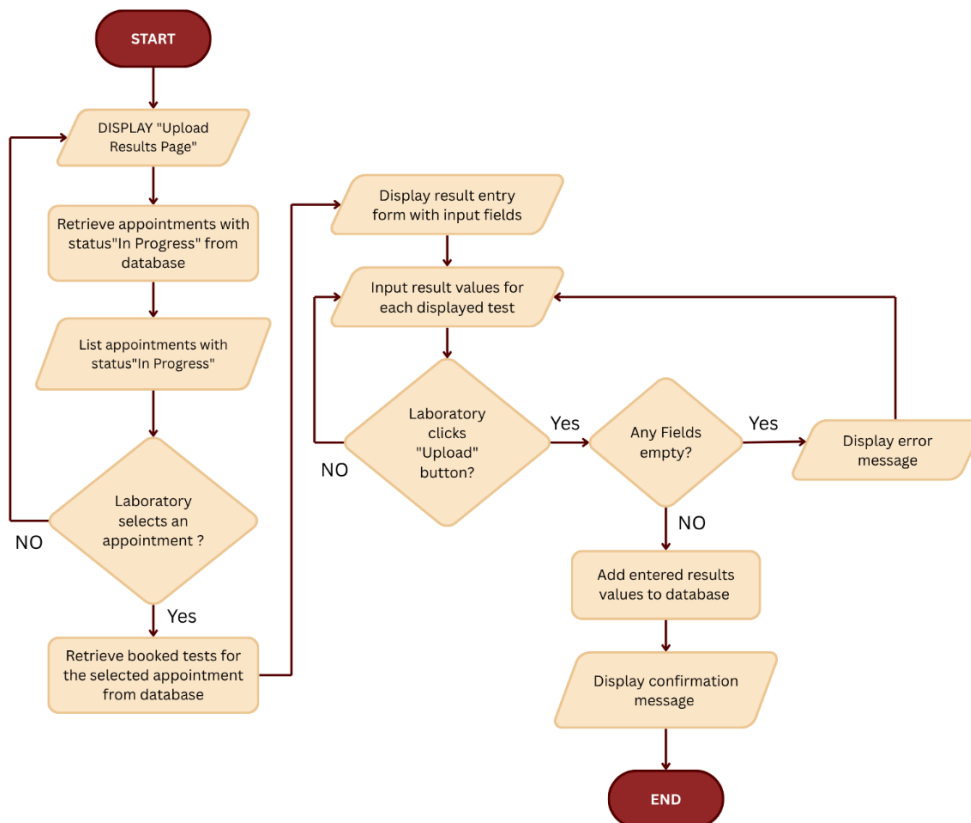


Figure 14- Upload Results Flowchart

- Block reported customers (Admin Dashboard)

The admin can manage customers reported by laboratories. The system displays a list of reported customers along with their details and report reasons. The admin can review each report and take action by blocking the user. Once blocked, the user is restricted from accessing system functionalities such as booking appointments.

Pseudocode:

BEGIN

DISPLAY "Reports Page"

RETRIEVE reported users from database

DISPLAY list of reported users

IF admin clicks on "Block" button for a selected user THEN

Update user status to "Blocked" in database

DISPLAY message to confirm the user is successfully blocked.

ENDIF

END

Flowchart:

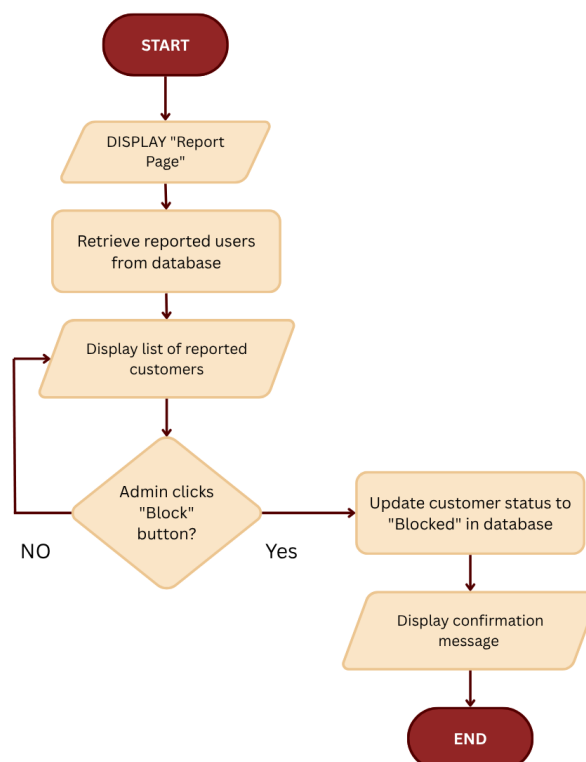


Figure 15- Block Flowchart

4.5 Interface Design

This section presents the interface design of the system through a site map that illustrates the overall structure of the application and the relationship between its main pages.

The system is designed to support three types of users: Customer, Laboratory, and Admin. Each user is directed to a specific dashboard after logging in, where they can access the functionalities relevant to their role. The site map provides a clear overview of how users navigate between pages and how the system is organized.

This structure ensures that the application is easy to navigate, user-friendly, and logically organized, allowing each user to efficiently perform their tasks within the system.

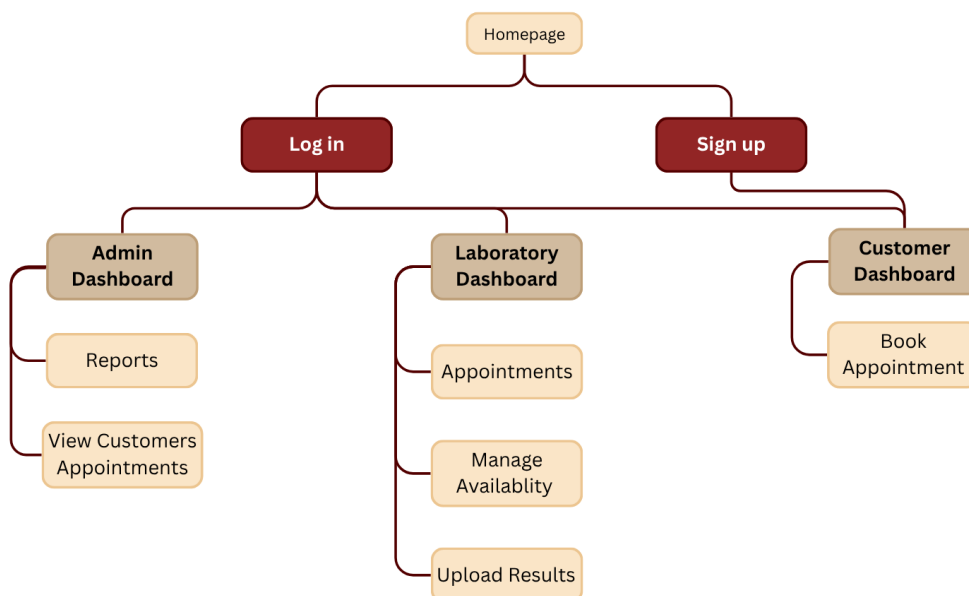


Figure 16- Nar'ak Sitemap

GitHub Link:

<https://github.com/MISSNOORA/NARAK>

Jira Link:

5 Chapter 5: Testing

5.1 User Story Acceptance Testing

User Story Acceptance Testing was conducted to ensure that each implemented feature in the NARAK platform aligns with its corresponding user story and satisfies the defined acceptance criteria. This testing approach focuses on validating system functionality from the user's perspective to ensure the application meets user expectations and requirements.

We first identified the user stories that represent the core features of the system, including managing appointment slots, booking laboratory tests, updating appointment statuses, and uploading test results. For each user story, specific test cases were designed and executed to verify whether the system behavior meets the expected outcomes.

The testing process was performed by simulating real user interactions within the system using different roles such as laboratory representatives and customers. Each user performed a set of tasks, such as adding time slots, booking appointments, and uploading results, while the system responses were observed and compared against the acceptance criteria.

The results of this testing were documented in the table below, including test outcomes and any relevant comments. The insights gained from this testing process helped identify minor issues, improve system functionality, and confirm that the system is ready for deployment.

Sprint Num	User Story	Acceptance Criteria	Test Cases	Pass?	Comments
2	As a customer, I want to register in the system so that I can create my personal account.	- As a customer, if any required field is empty, I should see an error message and the account should not be created. -As a customer, if I enter an invalid email	Test Case 1: Successful Registration 1-Go to the Welcome page and navigate to the registration page.	YES	

		format, the system should show an error message. -As a customer, if the email or phone number already exists, the system should show a message that the account already exists. -As a customer, when I type my password, the password characters should be masked (hidden). -As a customer, after successful registration, I should be redirected to my dashboard.	2-Enter a unique email, valid phone number, a strong password, and fill all required fields. 3-Click the "انشاء الحساب" button. 4-Verify that a success message appears and the user is redirected to their dashboard. Test Case 2: Invalid Registration Attempts 1-Try submitting the form with one or more required fields left empty — verify an error message appears: "الرجاء تعبئة جميع الحقول" 2-Enter an invalid email format verify the system shows an email format error message. 3-Enter an email or phone number that already exists in the system — verify the system shows: "هذا البريد الإلكتروني مسجل مسبقاً او رقم الجوال مستخدم مسبقاً."	YES	
--	--	--	---	-----	--

			4-Enter a password and verify that the characters are masked (hidden) while typing.		
2	As a NAR'AK user, I want to log in securely so that I can access my dashboard and services.	-As a user, if I enter correct email and password and click "Login", I should access my dashboard successfully. -As a user, if I enter incorrect email or password, I should see an error message. -As a user, if I leave email or password empty, I should see an error message. -As a user, when I type my password, the characters should be masked (hidden).	Test Case 1: Successful Login 1-Open the login page. 2-Select account type. 3-Enter a valid email and correct password. 4-Click "تسجيل الدخول". 5-Verify that the user is redirected to the dashboard and can see their information (name, appointments, etc.). Test Case 2: Invalid Login 1-Open the login page. 2-Enter an incorrect email or password. 3-Click "تسجيل الدخول". 4-Verify that an error message appears. Test Case 3: Empty Fields 1-Leave email or password empty. 2-Click "تسجيل الدخول".	YES	
				YES	

52

			redirected back to the login page.		
2	As a customer, I want to browse available laboratories so that I can see which labs and tests are available in the platform.	-As a customer, when I view my dashboard, I should see a list of available laboratories and their available test types. -As a customer, each laboratory should display its basic information.	Test Case 1: View Laboratories List 1. Log in/ Register to the system. 2. View the “available laboratories” section. 3. Verify that a list of laboratories is displayed. 4. Verify that each laboratory shows its basic information (name, test types, location, services)	YES	
2	As a customer, I want to search for a test type so that I can find what I need quickly.	-As a customer, if I type a test name in the search bar, I should see matching results. -As a customer, if no matching test is found, I should see a “No results found” message.	Test Case 1: Successful Test Search 1. Log in /Register to the system as a customer. 2. Go to the available laboratories section. 3. Type a valid test name in the search bar, such as “حديد”. 4. Verify that laboratories offering the matching test are displayed.	YES	

			Test Case 2: No Matching Results 1. Log in/Register to the system as a customer. 2. Type a test name that is not available in the search bar. 3. Verify that the system displays a “ لا توجد نتائج ” message.	YES	
2	As a customer, I want to select a laboratory so that I can book an appointment.	-As a customer, I should be able to select one laboratory. -As a customer, after selecting a laboratory, I should see the available test types and time slots.	Test Case 1: Successful Laboratory Selection 1. Log in/Register to the system as a customer. 2. Browse the list of available laboratories. 3. Click "حجز موعد" (Book Appointment) on one laboratory. 4. Verify that the booking form opens and displays the selected laboratory's available test types and time slots.	YES	
2	As a customer, I want to select the type of test so that I can	-As a customer, after selecting a laboratory, I should see the available test types.	Test Case 1: Successful Test Type Selection	YES	

	request the correct medical examination.	-As a customer, I should be able to select multiple test types. -As a customer, I should not continue booking without selecting a test type.	1-Open the booking form for a selected laboratory. 2-Verify that the available test types for that lab are displayed. 3-Select between 1 and 3 test types using the checkboxes. 4-Verify that the total price updates correctly based on selected tests. Test Case 2: Attempting to Continue Without Selecting a Test Type 1-Open the booking form for a selected laboratory. 2-Leave all test type checkboxes unchecked. Click "تأكيد الحجز". 3-Verify that the system does not proceed and shows an error message requiring at least one test to be selected.	YES	
--	--	--	---	-----	--

2	As a customer, I want to select an available time slot so that I can schedule my appointment at a convenient time.	-As a customer, after selecting a lab, I should see the available time slots. -As a customer, I should be able to select one available time slot. -As a customer, I should not continue booking without selecting a time slot.	Test Case 1: Successful Time Slot Selection 1-Open the booking form and select at least one test type. 2-Choose an available date from the date picker. 3-Select an available time from the time dropdown. 4-Click "مراجعة الحجز" then click "تعديل" or "تأكيد الحجز" and then verify the appointment is confirmed successfully. Test Case 2: Attempting to Confirm Without Selecting a Time Slot 1-Open the booking form and select at least one test type. 2-Leave the date or time fields empty.	YES	

			3-Click "تأكيد الحجز". 4-Verify that the system does not proceed and shows an error: "All fields are required! Please fill in the missing fields."	YES	
2	As a customer, I want to view my appointment details so that I can confirm the booking information.	-As a customer, after selecting the lab, test type, and time slot, I should see an appointment summary page. -As a customer, I should be able to confirm the booking only after viewing the summary. -As a customer, if I click "Confirm", the appointment should appear in my appointments list successfully. -As a customer, after successful booking, I should see a confirmation message. -As a customer, if I click "Cancel" on the	Test Case 1: Successful Appointment Confirmation 1. Log in/Register to the system as a customer. 2. From the dashboard, choose an available laboratory. 3. Click "حجز موعد". 4. Select the required test type(s), date, and time. 5. Click "مراجعة الحجز". 6. Verify that the appointment summary page displays the selected lab, tests, date, time, and total price. 7. Click "تأكيد الحجز".	YES	

		summary, the booking should not be created.	8. Verify that a confirmation message appears. 9. Verify that the appointment is added to the upcoming appointments list.		
2	As a customer, I want to edit my appointment so that I can change the time or test if needed.	-As a customer, if I click “تعديل”, I should be able to change the test type or time slot. -As a customer, if I select a new available slot and save, the appointment should update successfully. -As a customer, after editing, I should see a confirmation message	Test Case 1: Successful Appointment Edit 1. Log in to the system as a customer. 2. Go to the upcoming appointments list. 3. Click “تعديل” on an existing appointment. 4. Change the test type, date, or time slot. 5. Click “مراجعة الحجز”. 6. Verify that the updated appointment summary displays the new details. 7. Click “تأكيد الحجز”. 8. Verify that a confirmation message appears and the appointment details are updated in the appointments list.	YES	

			Test Case 2: Edit Without Saving 1. Log in to the system as a customer. 2. Go to the upcoming appointments list. 3. Click “تعديل” on an existing appointment. 4. Change the test type, date, or time slot. 5. Click “رجوع” or close the edit window without confirming. 6. Verify that the original appointment details remain unchanged.	YES	
2	As a customer, I want to cancel/delete my appointment so that I can remove it if I no longer need it.	-As a customer, if I click “Cancel”, I should be asked to confirm the cancellation. -As a customer, if I confirm, the appointment status should change to “Cancelled”. -As a customer, after cancellation, I should see a confirmation message.	Test case 1: cancel successfully 1-Navigate to the المواعيد القادمة 2-click “الغاء” 3-recvie popup message to confirm the cancellation 4- refresh page and navigate to the cancelled appointment section to view the cancelled appointment.	YES	

2	As a customer, I want to track the status of my test request so that I know the progress of my appointment.	-As a customer, when I view my appointment details, I should see the current status of the appointment. -As a customer, the status should clearly indicate whether it is Pending, Confirmed, In Progress, Completed, or Cancelled. -As a customer, if the laboratory updates the status, I should see the updated status when I refresh or reopen the page.	Test Case 1: Appointment Status Displayed on Dashboard 1. Log in to the system as a customer. 2. Navigate to the "المواعيد القادمة" section. 3. Verify that the current status of each appointment is clearly displayed (e.g., "في الانتظار") Test Case 2: Status Updates After Laboratory Changes It 1. Log in as a laboratory and update the status of a customer's appointment (e.g., from "Pending" to "Confirmed"). 2. Log in as the customer and go to the dashboard.	YES	
				YES	

			3. Verify that the updated status is correctly reflected directly on the dashboard.		
2	As a customer, I want to access all my laboratory reports in one account so that I can manage my medical history.	-As a customer, when I navigate to the Reports section, I should see a list of all my previous reports. -As a customer, if I click on a report, it should open successfully.	Test Case 1: Viewing All Reports in the Reports Section 1. Log in and navigate to the "Reports" section on the dashboard. 2. Verify that a list of all previous laboratory reports is displayed. Test Case 2: displaying Reports Successfully 1- Log in to the customer dashboard 2- navigate to the "نتائج الفحوصات" section. 3- all completed test results should be displayed	YES YES	

2	As a customer, I want to view my test results in a clear and understandable format so that I can easily read them	-As a customer, when I open a report, the test name and date should be clearly displayed. -As a customer, the result values should be clearly labeled. -As a customer, the results should not appear in a confusing or unorganized format.	Test Case 1: Test Name and Date Clearly Displayed 1.Log in and navigate to the "المواعيد السابقة" section. 2.Click on a completed appointment to expand it. 3.Verify that the test name and date are clearly displayed at the top of the report. Test Case 2: Result Values Are Clearly Labeled 1.Open a completed appointment report. 2.Verify that each test result shows the test name, result value (e.g., 12 nmol/L), unit, and status (e.g., "منخفض" / Low or	YES	
				YES	

			"طبيعي" / Normal) in a clearly labeled format. Test Case 3: Results Are Organized and Not Confusing 1.Open a completed appointment with multiple test results (e.g., Vitamin D, Vitamin B12, Calcium). 2.Verify that each test result is displayed in its own separate card. 3.Verify that the layout is clean, organized, and easy to read with no overlapping or missing information.		
				YES	

2	As a customer, I want to view a comparison of my current and previous test results so that I can monitor	-As a customer, when a new test result is available, the system should automatically compare it with my previous result of the same test type. -As a customer, I should see both the	Test Case 1: Automatic Comparison When New Result Is Available 1. Log in and navigate to the "نتائج الفحوصات" section.	YES	

	changes over time.	latest and previous result displayed clearly. -As a customer, the difference between the results should be visible.	2.Verify that when a new test result is available, the system automatically displays it alongside the previous result of the same test type (e.g., Vitamin D: previous 12 nmol/L vs current 18 nmol/L). Test Case 2: Latest and Previous Results Displayed Clearly 1.Navigate to the "نتائج الفحوصات" section. 2.Verify that for each test, both the current result and the previous result are clearly shown in separate labeled columns. 3.Verify that the normal range is also displayed for reference. Test Case 3: Difference and Trend Between Results Is Visible	YES	
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			1.Navigate to the " نتائج "الفحوصات section. 2.Click "مقارنة" (Compare) on any test (e.g., Vitamin D). 3.Verify that the trend chart shows the progression of results over time with values and dates clearly labeled. 4.Verify that the direction of change (e.g., ↑ ارتفع / ↓ انخفض / ثابت) is visible next to each result.	YES	
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2	As a laboratory representative, I want to add available appointment slots monthly so that customers can book home sample collection at convenient times.	-As a laboratory representative, if I open the availability section, I should be able to select available dates using a calendar interface instead of entering the date manually. -As a laboratory representative, if I attempt to select a past date from the calendar, I should be prevented from doing so. -As a laboratory representative, if I successfully add a date and time slot, it should appear immediately in the system and be visible to users for booking.	Test Case 1: Add Available Time Slot Successfully 1- Log in as a laboratory representative. 2- Go to “إدارة الأوقات”. 3- Select a date from the calendar. 4- View available time slots for the selected date. 5- Click “تفعيل” to make a time slot available. Test Case 2: Slot Becomes Visible for Booking 1- Log in as a laboratory representative. 2- Go to “إدارة الأوقات”. 3- Add a new valid future date and time	YES YES	

			slot. 4- Log out from the laboratory account. 5- Log in as a customer. 6- Go to the booking section. 7- Verify that the newly added slot is visible and available for booking.		
2	As a laboratory representative, I want to receive test appointments selected by customers so that I can prepare for sample collection and manage my schedule efficiently.	-As a laboratory representative, if a customer books an appointment using one of my available slots, I should see the appointment appear in my laboratory appointment list. -As a laboratory representative, if an appointment is booked by a customer, the selected slot should automatically become unavailable and should not be displayed as available to other customers.	Test Case 1: Appointment Appears in Dashboard 1- Log in as a laboratory. 2- Go to “طلبات المواعيد”. 3- View the list of appointments. 4- Verify that the booked appointment appears with customer details, test type, date, and time. Test Case 2: Slot Becomes Unavailable After Booking 1- Book an appointment as a customer.	YES YES	

		-As a laboratory representative, if I open an appointment details page, I should see the booking information including customer details, test type, appointment date, and appointment time. -As a laboratory representative, if there are no booked appointments, I should see a clear message indicating that there are currently no appointments.	2- Log in as a laboratory. 3- Go to “إدارة الأوقات”. 4- Verify that the booked slot is no longer available for selection.		
2	As a laboratory representative, I want to upload laboratory reports within the system so that customers can access their test results from their personal accounts.	-As a laboratory representative, if I select a specific booked appointment and open the results form, I should be able to enter the test results for that appointment. -As a laboratory representative, if I submit the results form with all required fields filled correctly, the results should be saved	Test Case 1: Upload Results Successfully 1- Log in as a laboratory. 2- Go to “طلبات المواعيد”. 3- Change the appointment status to “مكتمل”. 4- Go to “رفع النتائج”. 5- Select the completed appointment from the pending results list.	YES	

		successfully and linked to the selected appointment. -As a laboratory representative, if I leave any required result field empty and submit, I should see an error message asking me to fill in all required fields. -As a laboratory representative, if I enter a numeric test result in a text-only format when the test requires a number, I should see an error message indicating that the result must be numeric. -As a laboratory representative, if I save the results successfully, the appointment status should automatically update to "Completed".	6- 6- Enter the test result value . 7- Click "رفع النتائج". 8- Verify that the results are saved successfully and linked to the selected appointment by checking that: - The appointment no longer appears in the pending results list. - The result value is stored correctly in the system. Test Case 2: Submit Form with Missing Required Fields 1- Select an appointment from the pending results list. 2- Leave the result value field empty. 3- Click "رفع النتائج". 4- Verify that the system prevents		
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			submitting the form until the result value is entered.	YES	
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2	As a laboratory representative, I want to update the progress status of appointment requests so that customers can track the current stage of their laboratory test.	-As a laboratory representative, if I open an appointment request, I should be able to update its status to one of the allowed progress stages (e.g., Pending, Confirmed, In Progress, Completed, or Cancelled.). -As a laboratory representative, if I update the appointment status and save it, the new status should be stored successfully and reflected immediately in the laboratory dashboard. -As a laboratory representative, if I update the appointment status, the customer should be able to view the updated status in their	Test Case 1: Update Status Successfully 1- Log in as a laboratory. 2- Go to “طلبات المواعيد”. 3- Choose any appointment from the list. 4- Change the status from the dropdown menu. 5- Verify that the status updates immediately in the dashboard. Test Case 2: Customer Views Updated Appointment Status 1- Log in as a laboratory. 2- Go to “طلبات المواعيد”. 3- Choose any appointment from the list. 4- Change the	YES	
				YES	

		account when tracking the request.	appointment status (e.g., Confirmed or Completed). 5- Log out from the laboratory account. 6- Log in as the customer who booked the same appointment. 7- Go to the customer home page. 8- Check the “المواعيد القادمة” section. 9- Verify that the updated status is displayed correctly.		
2	As an admin, I want to view and manage customer and laboratory accounts so that I can maintain platform integrity and ensure only verified accounts operate on the system.	-As an admin, when I view my dashboard, I should see a list of latest appointments, reported customers and laboratories. -As an admin, I can delete/add a laboratory account. -As an admin, I should be able to block reported customer account. -As an admin, if I block a customer account, the customer	Test Case 1: Admin Dashboard Overview 1-Log in as an admin and go to Admin Dashboard. 2-Verify that the dashboard displays a list of latest appointments, reported customers, and laboratory accounts. Test Case 2: Add a New Laboratory Account	YES	

		should not be able to log in.	1-On the admin dashboard, locate the "إضافة مختبر جديد" form. 2-Fill in all required fields (lab name, email, phone, city, logo, tests offered with prices). 3-Submit the form and verify that the new laboratory appears in the "حسابات المختبرات" Test Case 3: Delete a Laboratory Account 1-On the admin dashboard, locate an existing laboratory in the "حسابات المختبرات" section. 2-Click the "حذف" button next to it. 3-Verify that the laboratory is removed from the list. Test Case 4: Block a Reported Customer Account 1-On the admin dashboard, locate a reported customer in	YES	
				YES	

			the "ملخص البلاغات" section. 2-Click to block the reported customer account. 3-Verify that the customer account is blocked. 4-Attempt to log in as that blocked customer and verify that login is denied.	YES	
2	As an admin, I want to monitor all laboratory test requests and view their detailed information so that I can ensure smooth workflow and	-As an admin, when I view the appointment section in my dashboard, I should see a list of all laboratory test requests. -As an admin, if I click on a specific appointment, I should be able to view its detailed information.	Test Case 1: Viewing All Test Requests on Dashboard 1-Log in as an admin and navigate to the appointment section on "الوحة المراقبة". 2-Verify that a list of all laboratory test requests is displayed with key details	YES	

	resolve issues when needed.	-As an admin, if no appointment exist, I should see a clear message indicating that there are no appointment available. -As an admin, if a laboratory updates the appointment status, I should see the updated status immediately when refreshing or reopening the page. -As an admin, I should be able to identify delayed or pending appointment based on their status and appointment date.	(customer name, lab, test, status). Test Case 2: Viewing Detailed Information of a Specific Appointment 1-From the appointment list, click on a specific appointment. 2-Verify that the full detailed information of that appointment is displayed (customer, lab, test type, date, status). Test Case 3: No Requests Available Message 1-Log in as an admin when no test requests exist in the system. 2-Navigate to the appointment section. 3-Verify that a clear message is shown indicating there are no appointments available.	YES YES YES	
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			Test Case 4: Status Updates After Laboratory Changes a Request 1-Log in as a laboratory and update the status of a test appointment. 2-Log in as an admin and navigate to the appointments section. 3-Verify that the updated status is immediately reflected.		
2	As NAR'AK users, I want the interface to be simple and easy to understand so that I can navigate the system without confusion.	- As a Nar'ak user, if I access my dashboard, I should clearly understand the main purpose of the platform within a few seconds. - As a Nar'ak user, if I navigate between pages, I should find consistent layout, colors, and button styles throughout the system. - As a Nar'ak user, if I click on any button or link, I should immediately	Test Case 1: Interface Clarity and Consistency 1. Log in to the system as a user. 2. Navigate through different pages (dashboard, booking, search). 3. Observe layout, colors, and button styles. 4. Verify that the design is consistent across all pages. 5. Verify that buttons and labels are clear	YES	The interface is simple, consistent, and easy to understand. Users can navigate the system easily, and error messages are clear and helpful.

		understand its function based on its label. - As a Nar'ak user, if I make an error while filling a form, I should see a clear and understandable error message.	and easy to understand. 6. Verify that the user can understand the purpose of the system within a few seconds. Test Case 2: Error Message Clarity 1. Go to a form (e.g., login or booking). 2. Enter invalid or incomplete data. 3. Submit the form. 4. Verify that clear and understandable error messages are displayed. 5. Verify that the user can easily understand how to fix the error.	YES	
2	As a Nar'ak user, I want the system to protect all personal and laboratory data so that my medical information remains secure	- As a Nar'ak user, I want my password to be stored securely so that it is not visible or stored in plain text within the system. - As a Nar'ak user, I want access to laboratory results and personal information to require	Test Case 1: Secure Authentication and Data Access 1. Open the system without logging in. 2. Try to access dashboard or any personal data page. 3. Verify that access is denied and the user is	YES	The system ensures secure authentication and proper access control. Sensitive data is protected and only accessible to

	and confidential.	authentication so that unauthorized individuals cannot view sensitive data. - As a Nar'ak user, I want the system to enforce proper access control so that only authorized roles can access specific information.	redirected to the login page. 4. Log in with valid credentials. 5. Verify that the user can access their personal data and laboratory results. Test Case 2: Access Control Validation 1. Log in as a customer. 2. Attempt to access data or pages that belong to another user or different role (e.g., admin/lab data). 3. Verify that access is restricted and not allowed. 4. Check that sensitive data is not exposed without proper authorization.	YES	authorized users.
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Table 8 - User Story Acceptance Testing

([Jira Sprint 2 complete](#))

5.2 Integration Testing

1. Integration hierarchy:

In this section, the integration hierarchy of the Nar'ak system is presented. The diagrams illustrate how the main system components are connected and interact with each other. Each diagram represents a specific user role in the system (customer, laboratory, and admin), showing how the functionalities are integrated within each role.

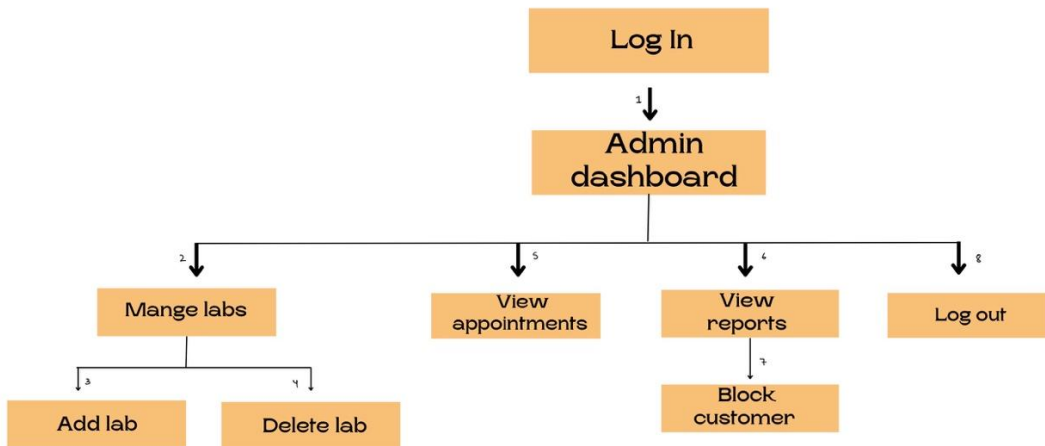


Figure 17 - Admin Integration hierarchy

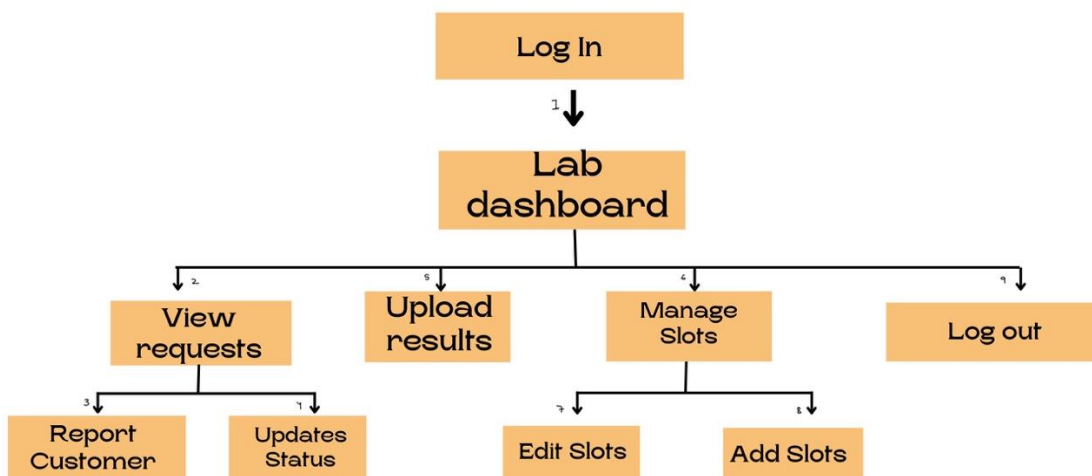


Figure 18 - Lab Integration hierarchy

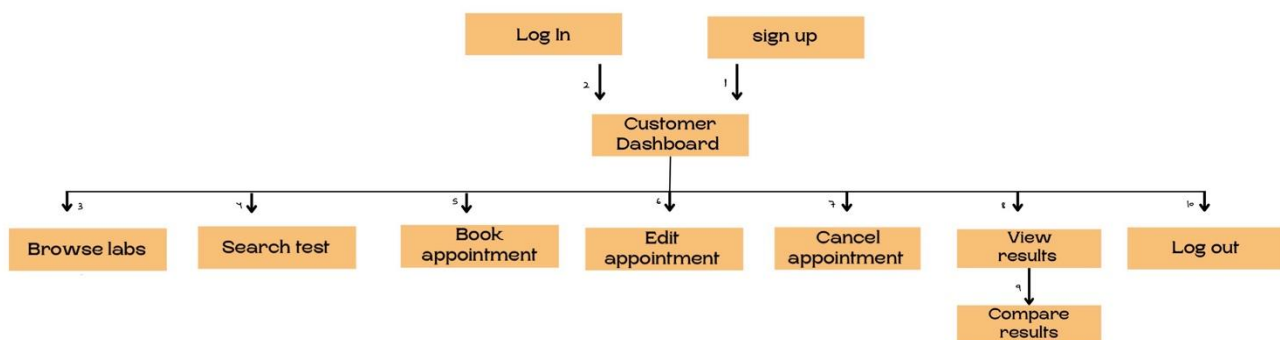


Figure 19 - Customer Integration hierarchy

2. Integration testing plan:

In this section, integration testing is performed to ensure that different system components in the Nar'ak system work correctly when integrated together. The testing focuses on the interaction between main modules such as login, booking appointments, and administrative actions.

Two main scenarios were selected to validate the system integration. The first scenario tests the customer booking process, while the second scenario tests the admin report handling process. These scenarios ensure that data flows correctly between system components and that the system behaves as expected.

Scenario 1: Customer books an appointment

System Components	New Component	Test Case	Pass?	Comments
- Log in to the system	-Browse available laboratories	1. A customer was logged in to the system successfully. 2. The customer dashboard displayed the available	Yes	-

		laboratories automatically.		
- Log in to the system -Browse available laboratories	-Booking appointment	1. A customer was logged in to the system successfully. 2. The customer dashboard displayed the available laboratories automatically. 3. The customer selected a laboratory from the displayed list. 4. The system displayed the selected laboratory information, available tests, prices, and time slots.	yes	

		5. The customer selected the required test type and available date and time. 6. The customer reviewed the booking summary and confirmed the appointment.		
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Table 9 - Integration Scenario 1

Scenario 2: Admin handles a reported customer

System Components	New Component	Test Case	Pass?	Comments
- Log in to the system	- Open reports section	1. The admin was logged in to the system successfully. 2. The admin clicked on the “Reports” button from the dashboard.	yes	

		3. The system displayed the list of reported customers with their details and report reasons.		
- Log in to the system - Open reports section	-Blocking reported customer	1. The admin was logged in to the system successfully. 2. The admin clicked on the “Reports” button from the dashboard. 3. The system displayed the list of reported customers with their details and report reasons. 4. The admin selected the “Block” option for a specific customer.	Yes	

		5. The system updated the customer status to “Blocked” and removed the report from the list.		
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Table 10 - Integration Scenario 2

5.3 User Acceptance Testing

5.3.1 Introduction

User Acceptance Testing (UAT) was conducted to evaluate whether the Nar’ak system meets user requirements and can be effectively used by the target audience. The purpose of this testing was to assess the system’s usability, functionality, and overall user experience based on real user interaction.

The testing involved five participants who match the target users identified in Phase 2. A structured questionnaire consisting of more than 10 well-designed and unbiased questions was used to collect feedback. The questions focused on system usability, task completion, clarity of interface, and usefulness of features.

5.3.2 Test Procedure

The User Acceptance Testing was conducted by providing participants with access to the Nar’ak system and asking them to interact with it in a realistic usage scenario.

Each participant was given a brief explanation of the system’s purpose. They were then asked to perform a set of key tasks, including registering/logging in, booking a laboratory appointment, selecting test types and time slots, viewing laboratory results, and comparing results.

Participants interacted with the system independently to simulate real user behavior. Minimal guidance was provided only when necessary to avoid confusion.

After completing the tasks, participants were asked to fill out a structured questionnaire created using Google Forms. The questionnaire included multiple-choice and open-ended questions focusing on ease of use, clarity of the interface, task completion, and usefulness of system features.

Finally, the responses were collected and analyzed, and the results were presented using charts and graphs to clearly illustrate user feedback.

5.3.3 Demographics of Participants

The demographic information of the participants is summarized as follows:

Gender:

The participants consisted of 60% females and 40% males.

نسخ الرسم البياني

?D2. What is your gender

ما هو جنسك؟

5 ردود

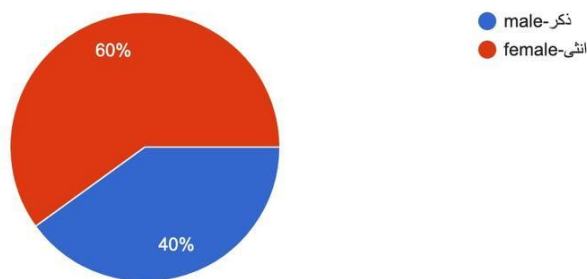


Figure 20 - Gender

Age Group:

40% of participants were aged between 18–25, 20% between 26–35, and 40% were aged 46 and above.

نسخ الرسم البياني

?D1. What is your age group

ما هو عمرك؟

5 ردود

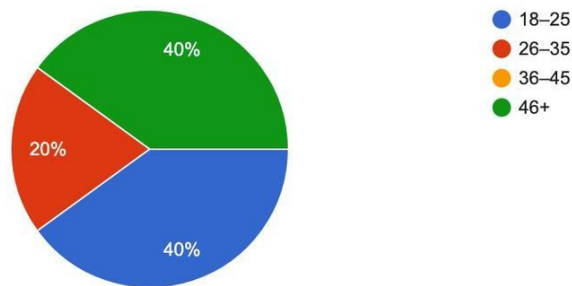


Figure 21 - Age Group

Occupation:

The participants included 40% students, 20% employees, and 40% unemployed individuals.

نسخ الرسم البياني

?D3. What is your occupation

ما هي وظيفتك؟

5 ردود

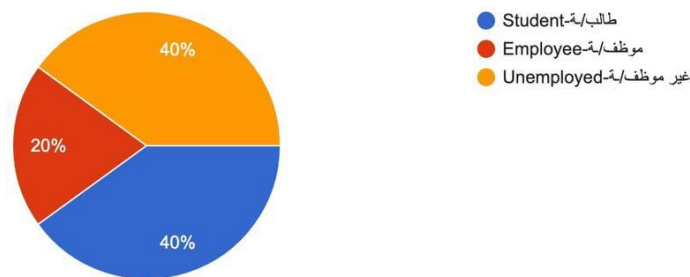


Figure 22 - Occupation

Technical Experience:

40% of participants had high technical experience, 20% had medium experience, and 40% had low technical experience.

نسخ الرسم البياني

?D4. How would you describe your level of experience with technology

كيف تصف مستوى خبرتك في استخدام التقنية؟

5 ردود

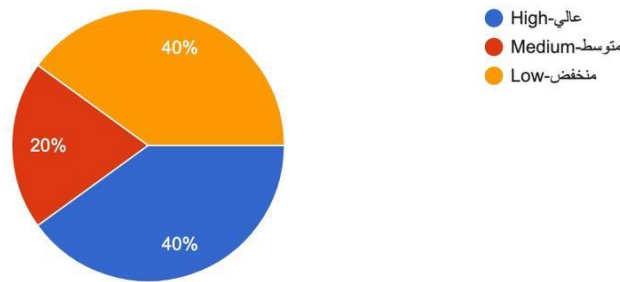


Figure 23 - Technical Experience

This diversity ensured that the system was evaluated by users with different backgrounds and levels of experience.

5.3.4 Results and Analysis

System Usability

The results show that the majority of participants (60%) were able to complete the booking process very easily without any problems, while 40% experienced minor issues. ([Appendix C Figure 32](#)) This indicates that the system is generally user-friendly and easy to use.

Additionally, 60% of participants found the system interface very clear, and 40% found it clear, which reflects a well-designed and understandable interface. ([Appendix C Figure 31](#))

Task Completion

All participants (100%) were able to successfully book an appointment and view laboratory results. In addition, 80% of participants were able to register/login and compare results successfully, while 60% were able to track appointment status. ([Appendix C Figure 27](#)). This demonstrates that the core functionalities of the system are effective and can be completed by most users.

Booking Efficiency

Regarding the time required to complete a booking, 60% of participants completed the process within 2–5 minutes, 20% in less than 2 minutes, and 20% within 5–10 minutes ([Appendix C Figure 26](#)). This indicates that the system allows efficient task completion. However, 80% of participants reported that selecting the time slot was the most time-consuming step, suggesting an area for improvement. ([Appendix C Figure 33](#))

Access to Results

The majority of participants (80%) were able to access their laboratory results easily from the dashboard, while 20% needed some time to find them ([Appendix C Figure 29](#)). This shows that the system provides effective access to results with minor usability improvements needed.

Feature Evaluation

When evaluating system features:

60% of participants selected home sample collection as the most useful feature.

40% selected result comparison, highlighting its importance for tracking health data([Appendix C Figure 28](#)).

However, 60% of participants reported that result comparison was the hardest feature to use, indicating a need for improvement in its design or presentation([Appendix C Figure 30](#)).

User Independence

Most participants (60%) were able to use the system independently without assistance, while 40% required minimal help([Appendix C Figure 25](#)). This indicates that the system is generally intuitive but may need slight improvements for first-time users.

Future Usage

A majority of participants (80%) stated that they would use the system again in the future, while 20% responded with “maybe.” ([Appendix C Figure 26](#)) This reflects a high level of user satisfaction and acceptance.

5.4 Discussion

The results of the system evaluation indicate that the Nar'ak platform performed well in terms of usability, efficiency, and user satisfaction. Based on the User Acceptance Testing (UAT), all participants were able to complete the main tasks, such as booking appointments and viewing laboratory results, which shows that the core functional requirements were successfully met.

In addition, 80% of users stated that they would use the system again, reflecting a high level of user satisfaction. Most users were also able to complete tasks within 2–5 minutes, indicating good system efficiency and alignment with non-functional requirements related to performance and usability.

From a usability perspective, 60% of users found the system very easy to use, and 60% were able to use it independently without assistance. Furthermore, 80% of users accessed their results easily from the dashboard, which suggests that the interface and navigation are generally clear and effective.

However, the evaluation identified areas for improvement. The result comparison feature was considered the most difficult to use by 60% of participants, indicating that its design or implementation may not be sufficiently intuitive. Additionally, only 60% of users were able to track appointment status successfully, suggesting limited visibility of this feature.

From a design perspective, the system is clear for basic tasks but requires improvement in more complex features. From a development perspective, all core functionalities were implemented successfully, but earlier usability testing could have improved certain features. Regarding the development process, applying more user-centered design practices and iterative testing could enhance overall system quality.

In terms of non-functional requirements (NFR), the system performed well in usability, performance, and accessibility. However, improvements are needed in learnability and user guidance, especially for first-time users.

Overall, the system evaluation results can be considered good, as the main objectives were achieved and users were able to use the system effectively. However, the following improvements are recommended:

- Simplifying the result comparison feature and adding visual aids
- Improving the visibility of appointment tracking
- Providing user guidance such as tooltips or instructions
- Conducting iterative usability testing during development

In conclusion, the Nar'ak system demonstrates strong performance in its core functionalities, but enhancing certain features and improving user support will further improve the overall user experience.

6 Chapter 6: Conclusion and Future Work

The Nar'ak platform was developed to address a real and widely experienced problem in Riyadh: the difficulty of accessing laboratory services efficiently. Many residents faced challenges including traffic congestion, fragmented communication with multiple laboratories, difficulty interpreting medical reports, and the inability to track and compare test results over time. By designing and building a unified, web-based platform, the Nar'ak team aimed to simplify this experience for users.

The development process followed an Agile Scrum methodology, progressing through multiple sprints that covered requirements elicitation, system design, implementation, and testing. The team conducted interviews and questionnaires to gather real user needs, designed a client-server architecture using PHP and MySQL, and implemented features spanning customer registration and login, laboratory browsing and booking, appointment tracking, result viewing, result comparison, and an administrative dashboard for managing accounts and monitoring activity.

6.1 Global and local impact.

At the local level, Nar'ak directly addresses a practical problem experienced by residents of Riyadh. The platform reduces the need for physical travel to laboratories by enabling home sample collection bookings, which is particularly valuable given the well-documented traffic congestion

in the city. By centralizing laboratory services into one accessible platform, Nar'ak saves users time, reduces stress, and increases the efficiency of the healthcare experience. Customers can now book appointments, track their status, receive results, and compare them over time, all within a single system. This contributes to raising health awareness by making it easier for individuals to monitor their health indicators regularly.

At the global level, Nar'ak reflects a broader movement toward digital health transformation. Across the world, healthcare systems are increasingly adopting technology to improve access to medical services, reduce administrative burden, and empower patients to manage their own health data. Nar'ak contributes to this direction by demonstrating how a lightweight, web-based platform can bridge the gap between patients and diagnostic services. While the current scope is limited to Riyadh, the core concept of a centralized laboratory booking and result management platform is applicable in any urban environment facing similar challenges of access and fragmentation.

6.2 Problems and challenges encountered during the software development

Throughout the development of Nar'ak, the team encountered several challenges that required careful problem-solving and adaptation.

One of the primary challenges was learning and applying PHP as a new programming language. Most team members had limited prior experience with server-side web development, which required additional time for learning and practice before meaningful progress could be made on the implementation.

Integrating the database with the web interface presented another significant challenge. Ensuring that data was correctly stored, retrieved, and displayed across different user roles required extensive debugging and testing, especially for features involving dynamic content such as appointment slots and test result comparisons.

Team coordination was also a challenge that the team had to manage carefully. Working on a shared codebase using GitHub required establishing clear conventions and consistent communication to avoid conflicts and duplication of work. Managing task distribution through Jira helped mitigate these issues, though alignment across team members still required ongoing effort.

6.3 Limitations of the system.

While Nar'ak successfully delivers its core functionality, several limitations exist in the current version of the system that should be acknowledged.

The platform is currently limited to the city of Riyadh and does not support laboratories or customers outside this geographic scope. It is also only accessible via desktop or laptop using the Google Chrome browser, which excludes mobile and tablet users and limits accessibility for a portion of the target audience.

The system supports Arabic language only, which while appropriate for the target audience in Riyadh, restricts use by non-Arabic speakers. Additionally, the platform does not integrate with any external health systems, insurance providers, or governmental health databases, meaning it operates as a standalone system without connection to wider healthcare infrastructure.

The system does not include an online payment feature, which means that financial transactions between customers and laboratories must be handled outside the platform. Furthermore, the system does not offer medical interpretation or clinical guidance for test results; it is designed for viewing and tracking purposes only, with no diagnostic functionality.

6.4 The main contribution of the project

The primary contribution of Nar'ak is the creation of a centralized, unified platform that connects customers with multiple certified laboratories in Riyadh. Prior to this platform, users had to contact laboratories individually through phone calls or messaging applications, manage results scattered across different formats and providers, and perform manual comparisons when tracking health changes over time.

Nar'ak consolidates this fragmented experience into a single web-based system where customers can register, browse available laboratories, book appointments with home visit support, track the progress of their requests, receive and view uploaded test results, and compare current results with previous ones visually. Laboratories benefit from a dedicated dashboard to manage appointments and upload results, while administrators can oversee the entire platform, manage laboratory accounts, and handle reported users.

The result comparison feature stands out as a unique contribution, as it allows users to track changes in their health indicators over time within the same platform, a capability not offered by existing laboratory service providers in Riyadh such as Dr. Sulaiman Al Habib and Al Borg Diagnostics. This empowers users to take a more proactive role in monitoring their own health.

6.5 Future work.

Looking ahead, several enhancements and expansions could significantly improve the Nar'ak platform and extend its reach and impact.

Mobile application support is one of the most impactful future developments. Making Nar'ak accessible on smartphones and tablets would greatly increase the platform's reach, as mobile devices are the primary means of internet access for a large portion of users in Saudi Arabia. Integrating an online payment system would complete the end-to-end booking experience, allowing customers to pay for their tests directly through the platform without requiring separate financial arrangements with laboratories. Adding a notification system that alerts customers via SMS or email when their appointment is confirmed, when their results are ready, or when a status update occurs would improve the responsiveness of the platform and reduce the need for customers to manually check the system.

Expanding the geographic scope beyond Riyadh to cover other cities in Saudi Arabia would increase the platform's impact and make it relevant to a much broader population. Multi-language support, particularly adding English, would also make the platform more inclusive and accessible to non-Arabic speakers residing in the country.

Finally, incorporating AI-based features such as simplified result interpretation, personalized health reminders, or trend analysis could further empower customers to understand their health data and act on it proactively. Such capabilities would elevate Nar'ak from a booking and tracking tool to a more comprehensive personal health management platform.

7 References

[1]Dr. Sulaiman Al Habib Medical Group, “Laboratory services,” 2024. [Online]. Available: <https://hmg.com/ar/pages/home.aspx>

[2]Al Borg Diagnostics, “Laboratory testing services,” 2024. [Online]. Available: <https://alborgdx.com>

[3] Google LLC, “Google Chrome web browser,” 2024. [Online]. Available: <https://www.google.com/chrome/>

8 Appendix

8.1 Appendix A – interview

8.1.1 interview questions

Question 1:

What difficulties do you face when booking laboratory tests?

Question 2:

Do you find laboratory test reports easy to understand? Why or why not?

Question 3:

Do you keep your previous laboratory test results? If yes, do you find it easy to access and compare them?

Question 4:

Have you ever faced difficulty reaching a laboratory due to traffic congestion or long distances?

Question 5:

How important is home sample collection to you? Why?

8.1.2 interview transcripts

Interview #1	
Interviewee: Male, 28 years old, Employee	Interviewer: Norah
Location/medium: Face to face interview in Riyadh	Appointment date: 30 Feb 2026 Start time: 09:02 End time: 09:44
Objectives: – To understand the challenges faced by people when booking laboratory tests. – To explore difficulties in understanding laboratory reports. – To evaluate the importance of home sample collection for users who repeat tests regularly.	Reminders: The participant prioritizes fast and flexible online booking options that fit around his busy work schedule.
Agenda: – Introduction – Overview of the project – Question 1 – Question 2 – Question 3 – Question 4 – Question 5 – Closing & additional comments	Approximate time: – 3 minutes – 4 minutes – 7 minutes – 6 minutes – 4 minutes – 7 minutes – 6 minutes – 5 minutes

Table 11- Interview 1 Transcript

Interview #1

General Observations:

The interviewee appeared professional and direct in his responses. He emphasized time management as the main concern due to his busy work schedule. He showed frustration regarding limited appointment slots and inconsistent laboratory formats. He values efficiency and prefers digital solutions that simplify the process.

Topic not covered: Data privacy policies, system security details, and technical integration with laboratory systems were not explored during this interview.

Interviewee: Male, 28 years old, Employee

Date: 30 Feb 2026

Questions:

Notes:

Answer:

Q1: What difficulties do you face when booking laboratory tests?

Because of my work schedule, timing is the main issue for me. Many laboratories only offer limited appointment windows, and they often fill up quickly. I usually have to adjust my working hours or request permission to attend. It's not easy to find a time slot that fits my day without affecting my responsibilities.

Observation:

The participant focused strongly on time constraints and the need for flexible scheduling.

Q2: Do you find laboratory test reports easy to understand? Why or why not?

Answer:

Not completely. I can understand basic numbers, but when it comes to detailed indicators or

	medical abbreviations, it becomes unclear. Sometimes values are slightly outside the normal range, and I'm not sure how serious that is. Without a medical background, it's difficult to interpret the meaning behind the numbers. Observation: The participant indicated a need for clearer explanations and simplified medical terminology.
Q3: Do you keep your previous laboratory test results? If yes, do you find it easy to access and compare them?	Answer: Yes, I keep them digitally. However, since each laboratory uses a different format, it's not straightforward to track changes over time. The layout and order of results vary, so comparing trends requires effort. Observation: The participant prefers centralized digital storage with consistent formatting.
Q4: Have you ever faced difficulty reaching a laboratory due to traffic congestion or long distances?	Answer: Yes, especially during peak hours. Leaving work and driving across the city can take longer than expected. There were times when I arrived late due to traffic, which added unnecessary stress to the day. Observation:

	Traffic congestion in Riyadh significantly impacts appointment convenience.
Q5: How important is home sample collection to you? Why?	Answer: It is important mainly because of time management. On busy weeks, going to a laboratory requires rearranging my schedule. Reducing travel time would make the process more manageable. Observation: Home sample collection is seen as a practical solution to time-related challenges.

Interview #2	
Interviewee: Female, 21 years old, University student	Interviewer: Yara
Location/medium: Online interview	Appointment date: 25 Feb 2026 Start time: 12:00 End time: 12:23
Objectives: – To understand the challenges faced by people when booking laboratory tests. – To explore difficulties in understanding laboratory reports.	Reminders: The participant is a university student who prefers simple and convenient processes due to her busy academic schedule

– To evaluate the importance of home sample collection for users who repeat tests regularly.	
Agenda:	Approximate time:
– Introduction	– 2 minutes
– Overview of the project	– 2 minutes
– Question 1	– 4 minutes
– Question 2	– 3 minutes
– Question 3	– 4 minutes
– Question 4	– 3 minutes
– Question 5	– 3 minutes
– Closing	– 2 minutes

Table 12- Interview 2 Transcript

Interview #2	
General Observations: The interviewee is a university student with a busy schedule. Time management and transportation are major challenges when booking laboratory appointments. She prefers flexible appointment times and convenient options such as home sample collection. She also struggles with understanding medical reports and comparing previous results, which indicates a need for clearer report presentation and automatic comparison features.	
Topic not covered: payment methods for booking laboratory appointments, preferred notification methods such as SMS or email. These areas may require further investigation in future interviews.	
Interviewee: Female, 21 years old, University student Date: 25 Feb 2026	
Questions:	Notes:
Q1: What difficulties do you face when booking laboratory tests?	Answer:

	Most of the available appointments are usually scheduled in the morning, which conflicts with my university classes. Because of that, I am forced to choose afternoon appointments. However, afternoons in Riyadh are typically during rush hours, which makes traffic very heavy. Reaching the laboratory can take much longer than expected, and it becomes stressful, especially after a long academic day filled with lectures and assignments. Sometimes, I even consider postponing the test because the process feels inconvenient. Observation: The participant faces scheduling conflicts and traffic issues, which makes booking inconvenient.
Q2: Do you find laboratory test reports easy to understand? Why or why not?	Answer: Not always. While some reports are simple and clear, others contain medical terms and technical indicators that I am not familiar with. It is difficult for me to understand whether a slight increase or decrease in certain values is something serious or normal. I often feel unsure about interpreting the results correctly without additional explanation from a doctor. Observation: Medical terminology reduces the clarity of laboratory reports for the participant.

Q3: Do you keep your previous laboratory test results? If yes, do you find it easy to access and compare them?	Answer: Yes, I usually keep my previous test results saved in my email or stored on my phone. However, I do not regularly review them because accessing older reports requires searching through different files. Comparing results manually takes time since I need to open multiple documents and check the values one by one. It would be much easier if they were organized in one place and compared automatically. Observation: The current method of storing reports does not support easy comparison
Q4: Have you ever faced difficulty reaching a laboratory due to traffic congestion or long distances?	Answer: Yes, especially during rush hours in Riyadh. Traffic congestion significantly increases travel time, which makes attending appointments inconvenient. Public transportation options are not always practical, and driving during busy hours can be exhausting. At one point, I considered using the metro, but it was overcrowded and uncomfortable. These transportation challenges sometimes make the entire process frustrating. Observation: Transportation and traffic congestion are major barriers for the participant. Accessibility and

	location-based convenience are important factors in improving the booking experience.
Q5: How important is home sample collection to you? Why?	Answer: Yes, especially during rush hours in Riyadh. Traffic congestion significantly increases travel time, which makes attending appointments inconvenient. Public transportation options are not always practical, and driving during busy hours can be exhausting. At one point, I considered using the metro, but it was overcrowded and uncomfortable. These transportation challenges sometimes make the entire process frustrating. Observation: Convenience is a key factor influencing the participant's preferences.

Interview #3	
Interviewee: Female, 45 years old, Mother	Interviewer: Farah
Location/medium: Face to face interview in Riyadh	Appointment date: 27 Feb 2026 Start time: 09:30 End time: 10:05
Objectives: To understand the challenges faced by people when booking laboratory tests.	Reminders: The participant is living in Riyadh who strongly dislikes traffic congestion and prefers time-saving and convenient healthcare services.

– To explore difficulties in understanding laboratory reports. – To evaluate the importance of home sample collection for users who repeat tests regularly.	
Agenda:	Approximate time:
– Introduction	– 3 minutes
– Overview of the project	– 3 minutes
– Question 1	– 6 minutes
– Question 2	– 4 minutes
– Question 3	– 5 minutes
– Question 4	– 6 minutes
– Question 5	– 5 minutes
– Closing	– 3 minutes

Table 13- Interview 3 Transcript

Interview #3	
General Observations: The interviewee appeared calm but expressed frustration when discussing traffic and appointment arrangements. She emphasized that the process of organizing laboratory tests feels exhausting due to traffic congestion and communication with multiple laboratories. She values comfort, simplicity, and time efficiency. Understanding medical terminology was also a concern, especially when results were slightly abnormal.	
Topic not covered: Online payment preferences and preferred notification methods (SMS/email) were not discussed and may require further investigation.	
Interviewee: Female, 45 years old, Mother	Date: 27 Feb 2026
Questions:	Notes:

Q1: What difficulties do you face when booking laboratory tests?	Answer: For me, the difficulty is mainly in organizing everything. The booking process often involves several steps, and it is not always clear what I should do first. It feels exhausting because I need to check multiple laboratories then wait for responses, and make sure the timing fits my daily routine. With household responsibilities, managing all these details feels overwhelming. I prefer a process that is simple and organized in one place without confusion. Observation: The participant struggles with process complexity and time organization also communication issues. She prefers a clear, structured, and centralized booking system that simplifies multiple steps into one platform.
Q2: Do you find laboratory test reports easy to understand? Why or why not?	Answer: Not really. I can understand the basic parts like the test type, but when I look at the detailed results, I feel lost. There are many symbols, percentages, and medical notes that are not explained clearly. Sometimes I search online to understand a term, but the information I find is not always simple or reassuring. I prefer when someone explains the results to me in plain language. Observation:

	The participant does not feel confident interpreting medical reports independently and prefers simplified explanations.
Q3: Do you keep your previous laboratory test results? If yes, do you find it easy to access and compare them?	Answer: Yes, I try to keep them because I repeat certain tests every few months. However, they are not arranged properly. Some are printed and stored in a drawer, and others are saved as files on my phone. When my doctor asks about previous results, I need time to gather everything. I cannot easily see changes over time unless I compare each report manually. Observation: The participant values long-term tracking of her health but lacks an organized system that supports quick comparison and easy retrieval.
Q4: Have you ever faced difficulty reaching a laboratory due to traffic congestion or long distances?	Answer: Yes, and it is one of the most frustrating parts for me. In Riyadh, traffic can suddenly become very heavy, even in the morning. What should take 20 minutes sometimes takes almost an hour. I also worry about being late for the appointment after waiting so long in traffic. By the time I arrive, I already feel stressed and tired. It makes the whole experience uncomfortable. Observation: Traffic congestion significantly increases emotional stress before appointments and

	negatively affects the overall healthcare experience.
Q5: How important is home sample collection to you? Why?	Answer: It is extremely important to me. As a mother, I already have daily responsibilities at home. If I can avoid driving long distances and sitting in waiting areas, it would make a big difference. Having the test done at home would save time, reduce stress, and allow me to continue my day normally. I would definitely prefer that option whenever possible. Observation: Home sample collection is strongly associated with comfort, time efficiency, and stress reduction for the participant.

Interview #4	
Interviewee: Female, 60 years old, Housewife	Interviewer: Noora
Location/medium: Face to face interview in Riyadh	Appointment date: 23 Feb 2026 Start time: 10:05 End time: 10:47
Objectives: – To understand the challenges faced by people when booking laboratory tests. – To explore difficulties in understanding laboratory reports.	Reminders: The participant performs blood tests regularly for health monitoring and prefers simple processes.

– To evaluate the importance of home sample collection for users who repeat tests regularly.	
Agenda:	Approximate time:
– Introduction	– 3 minutes
– Overview of the project	– 4 minutes
– Question 1	– 6 minutes
– Question 2	– 7 minutes
– Question 3	– 5 minutes
– Question 4	– 8 minutes
– Question 5	– 5 minutes
– Questions from the interviewee	– 4 minutes

Table 14- Interview 4 Transcript

Interview #4	
General Observations:	
The interviewee appeared calm and cooperative but expressed visible frustration when discussing traffic and appointment days. She emphasized emotional fatigue more than technical difficulty. She spoke openly about feeling tired from the process rather than the medical test itself.	
Topic not covered: Insurance integration and online payment methods were not discussed due to time constraints.	
Interviewee: Female, 60 years old, Housewife	Date: 23 Feb 2026
Questions:	Notes:
	Answer:
Q1: What difficulties do you face when booking laboratory tests?	Booking the test feels exhausting even before the appointment day. I usually need to contact more than one laboratory to check availability and

	prices. Sometimes they don't answer immediately, and I have to wait for confirmation. I also prefer morning appointments, but they are often fully booked. Because I have family responsibilities at home, I cannot always choose any time freely. It feels like too much effort just to organize a simple blood test. Observation: The participant focused on emotional fatigue and repeatedly described the process as "tiring."
Q2: Do you find laboratory test reports easy to understand? Why or why not?	Answer: No, I honestly find them confusing. The reports contain many numbers and medical abbreviations that I don't fully understand. Even when the value is slightly above or below normal, I am not sure whether it is serious or not. Sometimes I feel anxious looking at the results because I don't know how to interpret them correctly. That is why I usually prefer that a doctor explains them to me directly. Observation: The participant showed uncertainty and mild anxiety when discussing medical terminology.
Q3: Do you keep your previous laboratory test results? If yes, do you find it easy to access and compare them?	Answer: Yes, I keep my results because I repeat my tests regularly. However, they are not organized in one place. Some are saved on my phone, and others are printed and kept in a folder at home.

	When I try to review older results, it takes time to find them. Comparing numbers from different reports is not easy because each laboratory uses a different format. Observation: The participant values keeping records but struggles with organization and comparison.
Q4: Have you ever faced difficulty reaching a laboratory due to traffic congestion or long distances?	Answer: Yes, very often. Traffic in Riyadh can be very stressful, especially during peak hours. Sometimes I spend a long time sitting in the car before reaching the laboratory. Finding parking is another issue, and it adds more stress. On some days, I feel more tired from the drive than from the blood test itself. The whole experience becomes exhausting. Observation: Traffic and transportation were described as the most physically draining part of the experience.
Q5: How important is home sample collection to you? Why?	Answer: Home sample collection is very important to me because I repeat my blood tests regularly. Avoiding traffic and waiting rooms would reduce a lot of stress. It would save time and energy, especially on days when I already feel tired. Since I do these tests every few months, having someone come to my home would make

	the process much more comfortable and easier to manage.
	Observation:
	The participant strongly associated home sample collection with comfort and reduced stress.

8.2 Appendix B – Questionnaires

8.2.1 Questionnaires questions

Question 1: How do you usually book laboratory tests?

- a) Phone call
- b) WhatsApp / messaging
- c) Visiting the lab physically
- d) Online website/app

Question 2: What difficulties do you face when arranging laboratory tests?

- a) Comparing laboratories
- b) Finding available appointments
- c) Travel and traffic
- d) Communication with laboratories

Question 3: What information is most important when choosing a laboratory?

- a) Price
- b) Available appointment times
- c) Home sample collection availability
- d) Laboratory reputation

Question 4: How do you prefer to receive your laboratory results?

- a) Printed report
- b) PDF via WhatsApp/email
- c) Laboratory portal
- d) I collect them physically

Question 5: What feature would help you better manage your laboratory results?

- a) Storing all results in one place
- b) Comparing old and new results
- c) Downloading reports anytime
- d) Notifications when results are ready

8.3 Appendix C – UAT Questionnaires

8.3.1 Questionnaires Graphs results

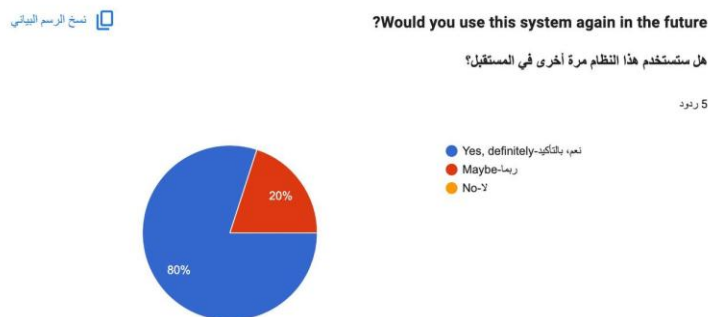


Figure 24

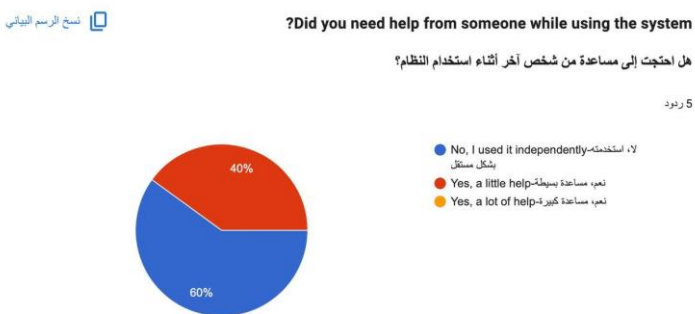


Figure 25

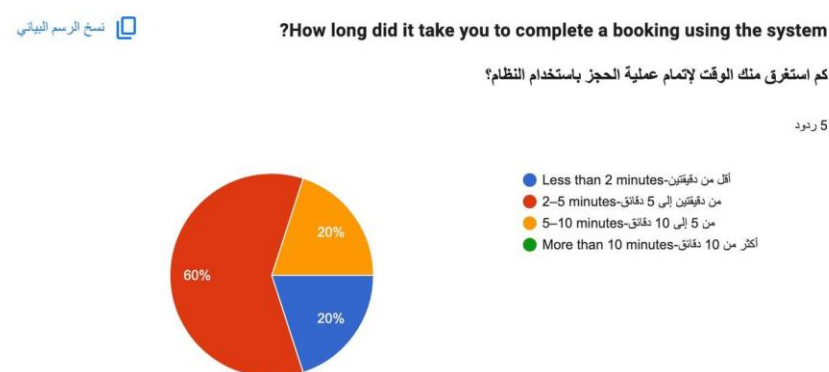


Figure 26

نسخ الرسم البياني

Which tasks were you able to complete successfully? (Select all that apply) .

ما هي المهام التي تمكنت من إنجازها بنجاح؟ (يمكن اختيار أكثر من خيار)

5 ردود

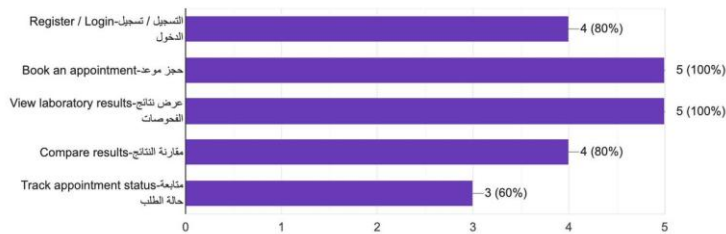


Figure 27

نسخ الرسم البياني

?Which feature did you find most useful in the system

ما هي الميزة الأكثر فائدة بالنسبة لك في النظام؟

5 ردود

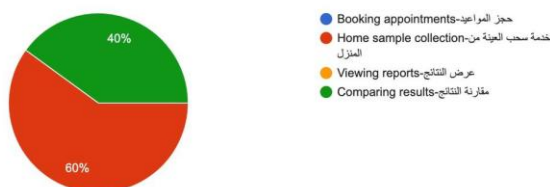


Figure 28

نسخ الرسم البياني

?How did you access your laboratory results

كيف تمكنت من الوصول إلى نتائج الفحوصات الخاصة بك؟

5 ردود

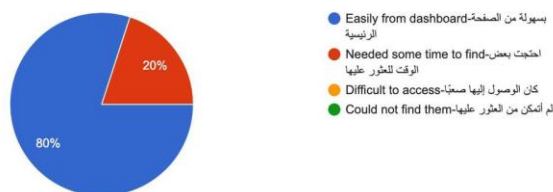


Figure 29

نسخ الرسم البياني

?Which feature was hardest to find or use

أي ميزة كانت الأصعب في العثور عليها أو استخدامها؟

5 ردود

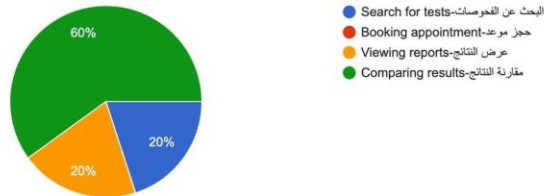


Figure 30

نسخ الرسم البياني

?How clear was the system interface (layout, buttons, pages)

ما مدى وضوح واجهة النظام (التصميم، الأزرار، الصفحات)؟

5 ردود

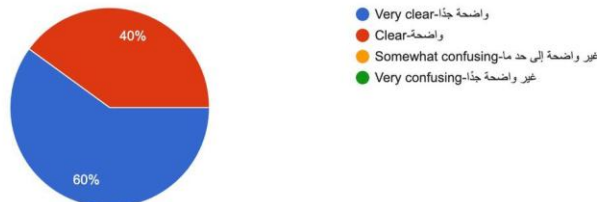


Figure 31

نسخ الرسم البياني

?How did you complete your booking in the system

كيف قمت بإتمام حجزك في النظام؟

5 ردود



Figure 32

نسخ الرسم البياني

?Which part of the booking process took you the most time

أي جزء من عملية الحجز استغرق منك وقتًا أطول؟

5 ردود

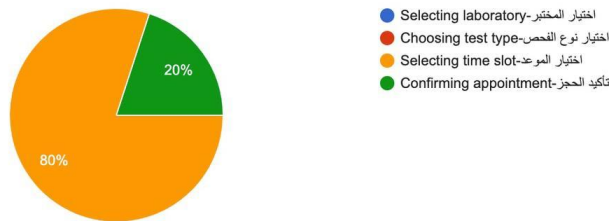


Figure 33

8.3.2 Questionnaires Questions

1. What is your age group?
2. What is your gender?
3. What is your occupation?
4. How would you describe your level of experience with technology?
5. Have you used laboratory services before?
6. How did you complete your booking in the system?
7. Which part of the booking process took you the most time?
8. How clear was the system interface (layout, buttons, pages)?
9. Which feature was hardest to find or use?
10. How did you access your laboratory results?
11. Which feature did you find most useful in the system
12. Which tasks were you able to complete successfully?
13. How long did it take you to complete a booking using the system?
14. Did you need help from someone while using the system?
15. Would you use this system again in the future?

Response Link:

8.3.3 Jira Screenshot (sprint 2):

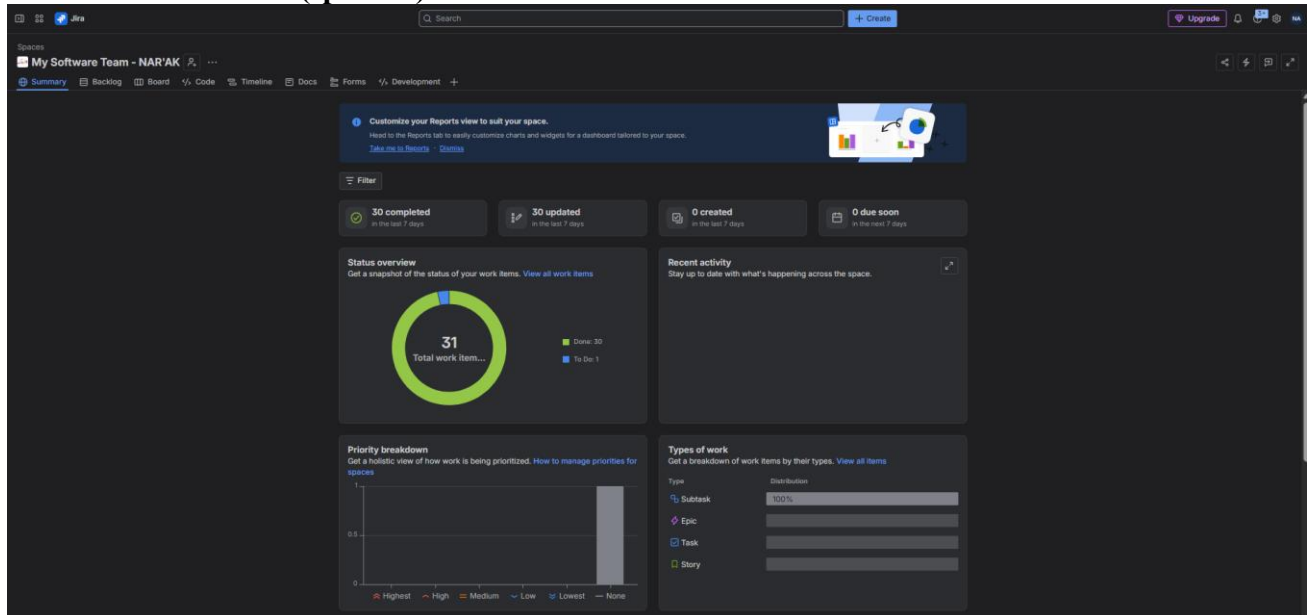


Figure 34-Jira Complete Sprint